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Temporal forecasting of $PM_{2.5}$, PM_{10}, NO_2 using deep learning models on Noida Sector 125 and Noida Sector 62

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Abstract

Air pollution has emerged as a critical environmental and public health issue in rapidly urbanising regions such as Noida, where pollutants including $PM_{2.5}$, PM_{10}, and NO_2 significantly deteriorate air quality and contribute to respiratory and cardiovascular diseases. Accurate forecasting of these pollutants is therefore essential for effective environmental monitoring and sustainable urban management. This study presents a comparative deep learning-based framework for multi-pollutant air quality forecasting at two major urban locations, Noida Sector 125 and Sector 62, representing institutional, commercial, and industrial environments with varying pollution characteristics. Time-series air quality datasets were collected from official monitoring agencies, including CPCB and SPCB, and subjected to preprocessing techniques such as missing value imputation, outlier handling, and Min-Max normalisation to ensure data reliability and consistency. Several deep learning architectures, including CNN, RNN, LSTM, GRU, and BiLSTM, were implemented to capture nonlinear temporal dependencies in pollutant concentration patterns. Furthermore, a hybrid BiLSTM-GRU model was proposed to improve forecasting performance by integrating bidirectional sequence learning with efficient gated recurrent mechanisms. The predictive performance of all models was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2). Experimental results demonstrated that the proposed Hybrid_BiLSTM_GRU model consistently outperformed the standalone models for all pollutants at both monitoring locations. At Noida Sector 62, the hybrid model achieved RMSE values of 41.01, 64.41, and 7.54 for $PM_{2.5}$, PM_{10}, and NO_2, respectively, along with corresponding R^2 scores of 0.4846, 0.4280, and 0.6391. Similarly, at Noida Sector 125, the model obtained RMSE values of 63.37, 90.05, and 16.90, with R^2 values of 0.2155, 0.3665, and 0.4748 for $PM_{2.5}$, PM_{10}, and NO_2, respectively. The hybrid framework also demonstrated significant percentage improvements over conventional deep learning models, achieving up to 43.08% reduction in MSE and 24.61% reduction in MAE for $PM_{2.5}$ prediction. Overall, the findings confirm that the proposed hybrid deep learning architecture effectively captures complex temporal pollution dynamics and provides reliable multi-pollutant forecasting for urban air quality management and environmental decision support systems.

Keywords: Air quality forecasting, Deep learning, BiLSTM-GRU, Time series prediction, PM2.5, PM10, Environmental monitoring

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1 Introduction

Air quality is a crucial indicator of environmental health and plays a significant role in determining the overall well-being of living organisms. It refers to the level of pollutants present in the atmosphere, which may include particulate matter such as $PM_{2.5}$ and PM_{10}, as well as gaseous pollutants like NO_2 [1]. These pollutants originate from both natural and anthropogenic sources. Natural sources include dust storms, wildfires, and volcanic eruptions, while human activities such as industrial emissions, vehicular exhaust, construction activities, and burning of fossil fuels are the primary contributors to air pollution in urban areas. In recent years, rapid urbanisation and industrialisation have significantly worsened air quality, particularly in developing countries [2]. Cities like Delhi frequently experience hazardous air pollution levels due to high population density, traffic congestion, and industrial activities. Seasonal factors such as temperature inversion, low wind speed, and crop residue burning further aggravate the problem by trapping pollutants near the ground. As a result, air pollution has become a major environmental and public health concern [3]. Exposure to poor air quality has serious health implications. Fine particulate matter $PM_{2.5}$ is especially dangerous because of its small size, which allows it to penetrate deep into the respiratory system and enter the bloodstream [4]. Prolonged exposure can lead to respiratory diseases such as asthma, bronchitis, and chronic obstructive pulmonary disease (COPD), as well as cardiovascular problems and premature death. Gaseous pollutants like NO_2 can cause irritation in the respiratory tract, while high concentrations of ozone can damage lung tissues. In addition to human health, air pollution negatively impacts vegetation, reduces agricultural productivity, and contributes to environmental issues such as acid rain and global warming [5]. Monitoring and analysis of air quality are essential for understanding pollution patterns and implementing effective control measures.

To address these challenges, advanced computational techniques such as machine learning and deep learning have gained significant attention in recent years. Deep learning models, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU), are capable of handling large and complex datasets [6]. These models can automatically learn patterns and relationships between multiple pollutants and meteorological variables, enabling more accurate and reliable air quality forecasting. In this study, a multi-pollutant approach is adopted to analyse and predict air quality using deep learning techniques. This integrated approach helps in capturing both spatial and temporal variations in air pollution. By leveraging hybrid deep learning models, the study aims to improve prediction accuracy and provide meaningful insights for environmental management [7]. Improving air quality is essential for ensuring sustainable development, protecting public health, and maintaining ecological balance [8].

Air quality assessment largely depends on the concentration of key pollutants present in the atmosphere. Among these, particulate matter ($PM_{2.5}$ and PM_{10}) and nitrogen dioxide (NO_2) are considered critical indicators due to their significant impact on human health and the environment [9]. These pollutants originate from various natural and anthropogenic sources and exhibit complex interactions with meteorological conditions. Particulate matter (PM) refers to a mixture of solid particles and liquid droplets suspended in the air. $PM_{2.5}$ consists of fine particles with a diameter of 2.5 micrometres

or less. Due to their extremely small size, these particles can penetrate deep into the respiratory tract and reach the alveoli of the lungs. $PM_{2.5}$ is primarily generated from combustion processes such as vehicle emissions, industrial activities, power plants, and biomass burning [10]. Secondary formation also occurs when gaseous pollutants react in the atmosphere to form fine particles. Exposure to $PM_{2.5}$ is associated with severe health issues, including respiratory infections, cardiovascular diseases, lung cancer, and premature mortality. It is considered one of the most harmful pollutants due to its ability to enter the bloodstream and affect multiple organs [11]. PM_{10} refers to coarse particulate matter with a diameter of 10 micrometres or less [12]. Prolonged exposure to PM_{10} can lead to coughing, throat irritation, and aggravation of existing respiratory conditions such as asthma and bronchitis. Both $PM_{2.5}$ and PM_{10} contribute to reduced visibility, commonly referred to as haze, and can negatively impact the climate by influencing the Earth's radiation balance. Nitrogen dioxide (NO_2) is a gaseous pollutant primarily produced from high-temperature combustion processes. Major sources of NO_2 include vehicular emissions, power plants, industrial activities, and the burning of fossil fuels. NO_2 plays a significant role in atmospheric chemistry as it contributes to the formation of ground-level ozone and secondary particulate matter. Exposure to elevated levels of NO_2 can irritate the respiratory system, reduce lung function, and increase susceptibility to respiratory infections [13]. It is particularly harmful to children, the elderly, and individuals with pre-existing respiratory conditions. In addition to health impacts, NO_2 contributes to environmental issues such as acid rain and the formation of photochemical smog [14].

The concentrations of $PM_{2.5}$, PM_{10}, and NO_2 are influenced by meteorological factors such as temperature, humidity, wind speed, and wind direction [15]. For instance, low wind speeds and temperature inversions can lead to the accumulation of pollutants near the ground, resulting in poor air quality conditions. Seasonal variations also play a crucial role, with higher pollution levels often observed during winter months due to stagnant atmospheric conditions and increased emissions from heating sources. Regulatory agencies establish permissible limits for these pollutants to ensure environmental safety. However, rapid urbanisation and industrial growth have made it challenging to maintain these standards, especially in metropolitan cities [16]. In recent years, advanced data-driven approaches such as machine learning and deep learning have been increasingly applied to analyse and predict the concentrations of $PM_{2.5}$, PM_{10}, and NO_2. [17] These models can capture complex nonlinear relationships between pollutants and meteorological variables, enabling accurate forecasting of air quality. Accurate prediction of these parameters helps in early warning systems, policy formulation, and implementation of effective pollution control strategies. In conclusion, $PM_{2.5}$, PM_{10}, and NO_2 are among the most critical air pollutants that significantly affect human health and environmental sustainability. Understanding their sources, behaviour, and impacts is essential for developing effective mitigation strategies and improving overall air quality [18].

Air pollution is a complex phenomenon that involves the interaction of multiple pollutants rather than the presence of a single contaminant. A multi-pollutant approach considers the combined effects of various air pollutants such as particulate matter $PM_{2.5}$ and PM_{10}, nitrogen dioxide (NO_2) [19]. This approach provides a more comprehensive understanding of air quality compared to single-pollutant analysis, as different

pollutants often interact with each other and with meteorological factors in the atmosphere. In urban environments, pollutants are emitted from multiple sources, including vehicular traffic, industrial processes, construction activities, and combustion of fossil fuels [20]. These emissions result in the simultaneous presence of various pollutants in the air. For example, nitrogen oxides and volatile organic compounds contribute to the formation of ground-level ozone through photochemical reactions, while sulfur dioxide and nitrogen dioxide can lead to the formation of secondary particulate matter. These interactions highlight the importance of studying pollutants collectively rather than in isolation. The multi-pollutant framework is particularly useful for understanding the cumulative impact of air pollution on human health. Exposure to a combination of pollutants can have synergistic effects, meaning that the combined impact may be greater than the sum of individual effects. Fine particulate matter $PM_{2.5}$ can carry toxic substances into the respiratory system, while gases like NO_2 can irritate airways and reduce lung function [21]. Long-term exposure to multiple pollutants has been linked to respiratory diseases, cardiovascular disorders, and increased mortality rates. For instance, low wind speeds can lead to the buildup of pollutants, while higher solar radiation promotes photochemical reactions that increase ozone formation. Therefore, multi-pollutant analysis often incorporates meteorological variables to improve the understanding of pollution dynamics.

With the advancement of computational techniques, data-driven approaches such as machine learning and deep learning have become increasingly popular for multi-pollutant analysis. Models such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and hybrid architectures are capable of capturing complex nonlinear relationships among multiple pollutants and environmental factors [22, 14]. These models enable accurate prediction of air quality by analysing large datasets collected from monitoring stations. The use of multi-pollutant models is particularly beneficial for air quality forecasting and decision-making. By considering multiple pollutants simultaneously, these models provide more reliable predictions and help in identifying pollution sources and trends [23]. This information is essential for policymakers and environmental agencies to design effective control strategies, implement regulations, and issue early warnings to the public. In conclusion, the multi-pollutant approach offers a holistic perspective on air quality analysis by considering the interactions among various pollutants and meteorological factors. It enhances the accuracy of prediction models and provides valuable insights into the complex nature of air pollution. Adopting this approach is crucial for effective environmental management and for mitigating the adverse impacts of air pollution on human health and ecosystems [24].

Deep learning is a subfield of machine learning that focuses on the use of artificial neural networks with multiple layers to model complex patterns in large datasets [25]. It is inspired by the structure and functioning of the human brain, where interconnected neurons process information in a hierarchical manner. Unlike traditional machine learning techniques, which often require manual feature extraction, deep learning models are capable of automatically learning features directly from raw data. This ability makes deep learning highly effective in handling complex, high-dimensional, and nonlinear data. Deep learning models are composed of multiple layers, including input layers,

hidden layers, and output layers [26]. Each layer consists of a set of neurons that apply mathematical transformations to the input data. The output of one layer serves as the input to the next layer, allowing the model to learn hierarchical representations of data. Activation functions such as sigmoid, hyperbolic tangent (tanh), and rectified linear unit (ReLU) introduce nonlinearity into the model, enabling it to capture complex relationships between variables [27].

Several types of deep learning architectures are commonly used for different types of data. Convolutional Neural Networks (CNNs) are widely used for processing spatial data and are effective in extracting features from images and grid-like data. Recurrent Neural Networks (RNNs) are designed for sequential data and are capable of capturing temporal dependencies [28]. Advanced variants such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) overcome the limitations of traditional RNNs and are particularly useful for time-series prediction tasks. Bidirectional LSTM (Bi-LSTM) further enhances performance by processing data in both forward and backward directions [29]. Deep learning has significant applications in environmental science, especially in air quality monitoring and forecasting. By integrating multiple pollutants and meteorological parameters, deep learning models can capture complex relationships and provide accurate predictions of air pollution levels. Hybrid models that combine different architectures, such as CNN-LSTM or BiLSTM-GRU, GRU-LSTM [30, 31] have shown improved performance by leveraging the strengths of multiple models.

Ongoing research is focused on improving model interpretability and efficiency. In conclusion, deep learning is a powerful tool for modelling complex systems and has become an essential technique in modern data analysis. Its ability to automatically learn features, handle large datasets, and model nonlinear relationships makes it highly suitable for air quality prediction and other real-world applications. With continuous advancements in technology and algorithms, deep learning is expected to play an increasingly important role in solving environmental challenges and improving decision-making processes [32].

2 Novelties

- A novel Hybrid_BiLSTM_GRU deep learning framework was proposed for multi-pollutant air quality forecasting, combining the bidirectional learning capability of BiLSTM with the efficient gating mechanism of GRU to capture complex temporal dependencies in air quality data.
- The study performed a comprehensive comparative analysis of multiple deep learning models, including CNN, RNN, LSTM, GRU, BiLSTM, and the proposed hybrid model, for predicting $PM_{2.5}$, PM_{10}, and NO_2 concentrations at two urban monitoring stations, namely Noida Sector 62 and Noida Sector 125.
- The proposed Hybrid_BiLSTM_GRU model achieved superior predictive performance with significant error reduction. At Noida Sector 62, the model achieved RMSE values of 41.01 for $PM_{2.5}$, 64.41 for PM_{10}, and 7.54 for NO_2, while providing up to 43.08% improvement in MSE and 24.61% improvement in MAE compared to conventional deep learning models.

- The research demonstrated the practical applicability of deep learning-based forecasting systems for urban environmental monitoring and smart city applications by integrating pollutant and meteorological parameters for reliable and location-specific air quality prediction.

3 Literature review

Air pollution has emerged as one of the most critical environmental challenges worldwide, significantly affecting human health, ecosystems, and climate systems [33]. Rapid urbanisation, industrialisation, and increased vehicular emissions have led to a substantial rise in atmospheric pollutants, particularly in developing countries. Various studies have emphasised the importance of monitoring key pollutants such as particulate matter $PM_{2.5}$ and PM_{10} and gaseous pollutants NO_2 due to their severe impact on human health and environmental sustainability. These pollutants originate from both natural sources, including dust storms and wildfires, and anthropogenic activities such as industrial emissions, construction work, and fossil fuel combustion [34]. Particulate matter, especially $PM_{2.5}$, is considered one of the most hazardous pollutants due to its ability to penetrate deep into the respiratory system and enter the bloodstream. Studies have linked prolonged exposure to $PM_{2.5}$ with respiratory diseases, cardiovascular disorders, and increased mortality rates [35].

Similarly, PM_{10}, although larger in size, contributes to respiratory irritation and reduced lung function. Nitrogen dioxide (NO_2), a byproduct of combustion processes, plays a crucial role in atmospheric chemistry by contributing to the formation of secondary pollutants such as ozone and particulate matter [36]. Exposure to NO_2 has been associated with respiratory infections, reduced lung capacity, and environmental issues such as acid rain and photochemical smog [37]. These findings highlight the need for comprehensive air quality assessment frameworks that consider multiple pollutants simultaneously. Traditional approaches to air quality analysis primarily relied on statistical and deterministic models [38]. While these methods provided foundational insights, they often failed to capture the complex, nonlinear relationships between pollutants and meteorological variables. Atmospheric processes are influenced by various factors such as temperature, humidity, wind speed, and solar radiation, which interact in a highly dynamic manner [39]r. For instance, low wind speeds and temperature inversions can lead to the accumulation of pollutants, whereas solar radiation promotes photochemical reactions that increase ozone formation. Such complexities necessitate the adoption of advanced modelling techniques capable of handling nonlinear and high-dimensional data.

In recent years, machine learning and deep learning techniques have gained significant attention for air quality prediction and analysis [40]. These data-driven approaches have demonstrated superior performance compared to traditional models due to their ability to learn complex patterns from large datasets. Models such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Random Forests have been widely applied for predicting pollutant concentrations [41]. However, deep learning architectures have shown even greater potential in capturing temporal and spatial dependencies in environmental data. Deep learning models, including Convolutional Neural Networks

(CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU), have been extensively explored in recent studies [42]. CNN models are effective in extracting spatial features, while RNN-based models are suitable for sequential data analysis. LSTM and GRU models address the limitations of traditional RNNs by incorporating memory mechanisms that enable them to capture long-term dependencies in time-series data [43]. These architectures have proven particularly useful for air quality forecasting, where pollutant concentrations vary over time and are influenced by multiple interacting factors.

Furthermore, hybrid deep learning models have been proposed to enhance prediction accuracy by combining the strengths of multiple architectures [44]. For instance, CNN-LSTM models integrate spatial feature extraction with temporal sequence modelling, while combinations such as BiLSTM-GRU improve robustness and generalisation. Studies have reported that hybrid models outperform standalone models in terms of accuracy and reliability, making them highly suitable for complex environmental prediction tasks [45]. Another significant advancement in air quality research is the adoption of a multi-pollutant approach. Unlike traditional single-pollutant analysis, this approach considers the combined effects and interactions of multiple pollutants. Research has shown that exposure to a mixture of pollutants can have synergistic effects, leading to more severe health outcomes compared to exposure to individual pollutants. Multi-pollutant models also provide a more realistic representation of environmental conditions, as pollutants often coexist and interact through chemical reactions in the atmosphere [46].

The integration of meteorological parameters into predictive models has further improved the accuracy of air quality forecasting. Factors such as temperature, humidity, wind direction, and solar radiation significantly influence pollutant dispersion and transformation. By incorporating these variables, deep learning models can better capture the dynamic behaviour of air pollution and provide more reliable predictions [47]. This integrated approach is essential for developing effective early warning systems and supporting policy decisions. Despite the advancements in deep learning, several challenges remain. These models require large amounts of high-quality data for training and are computationally intensive. Additionally, the lack of interpretability in deep learning models poses challenges for understanding their decision-making processes. Researchers are increasingly focusing on developing explainable artificial intelligence techniques to address these limitations and improve model transparency [48].

A significant shift from traditional statistical methods to advanced deep learning techniques for air quality prediction. The use of multi-pollutant approaches, hybrid models, and integration of meteorological factors has greatly enhanced the accuracy and reliability of air quality forecasting systems. These advancements provide valuable insights for environmental management, policy formulation, and public health protection. As air pollution continues to pose a global challenge, the development of robust and intelligent prediction models remains a critical area of research [22].

The flowchart presented in Fig. 1 illustrates the relationship between industrialisation, urbanisation, and their impact on air pollution, along with the pathways toward achieving Sustainable Development Goals (SDGs). Industrialisation and urbanisation act as primary drivers that lead to increased air pollution due to higher emissions from industries, transportation, and urban activities. This increase in air pollution results

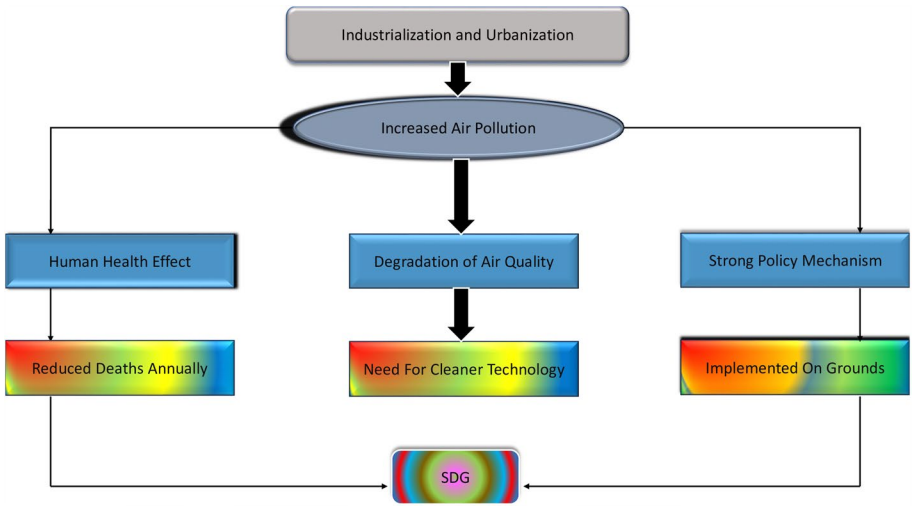


Fig. 1 Impact of industrialisation and urbanisation on air pollution and Sustainable Development Goals (SDGs)

in multiple consequences and response mechanisms. On one hand, it directly affects human health, leading to various respiratory and cardiovascular diseases; however, with appropriate interventions, this can contribute to a reduction in annual deaths. On the other hand, increased pollution leads to the degradation of air quality, which creates a demand for cleaner and more sustainable technologies. Additionally, it necessitates the development of strong policy mechanisms, where governments and institutions implement regulations and environmental policies on the ground. These three pathways, health improvements, technological advancements, and policy implementation, collectively contribute toward achieving the Sustainable Development Goals by promoting environmental sustainability, improving public health, and ensuring long-term ecological balance.

4 Data

The study focuses on two prominent locations in Noida, namely Sector 125 and Sector 62, which represent distinct urban environments with varying pollution characteristics. These areas are selected due to their significant contribution to urban air pollution and availability of continuous monitoring data for key pollutants such as $PM_{2.5}$, PM_{10}, and NO_2. Noida Sector 125 is a rapidly developing region characterised by institutional zones, commercial establishments, and proximity to major road networks. The presence of high traffic density and nearby construction activities contributes to elevated levels of particulate matter, particularly $PM_{2.5}$ and PM_{10}. Vehicular emissions are a major source of nitrogen dioxide (NO_2) in this area. Due to its urban setting and increasing population density, Sector 125 often experiences fluctuations in air quality, especially during peak traffic hours and seasonal changes. On the other hand, Noida Sector 62 is a well-established industrial and commercial hub, housing numerous IT companies, offices, and small-scale industries [49]. The area experiences high vehicular movement along with emissions from nearby industrial activities. As a result, concentrations of PM_{10} tend to be higher due to dust and industrial processes, while $PM_{2.5}$ and NO_2 levels

are influenced by traffic emissions and fuel combustion. The air quality in Sector 62 is also affected by the regional transport of pollutants from surrounding areas. Both locations are influenced by meteorological conditions such as wind speed, temperature, and humidity, which play a crucial role in the dispersion and accumulation of pollutants [50]. During low wind conditions, pollutants tend to accumulate, leading to poor air quality, whereas higher wind speeds facilitate dispersion. Seasonal factors such as winter inversion further worsen pollution levels in both regions. The selection of these two sites enables a comparative analysis of air quality in different urban settings, providing insights into the spatial variation of pollutant concentrations. By analysing $PM_{2.5}$, PM_{10}, and NO_2 levels in these areas, the study aims to understand pollution patterns and develop accurate prediction models using deep learning techniques. This helps in identifying pollution hotspots and supports the development of effective mitigation strategies [51].

4.1 Study area and data sources

The present study focuses on two important urban locations in Noida, namely Sector 125 and Sector 62, which are known for their distinct environmental and pollution characteristics [52]. These locations have been selected due to their high population density, rapid urban development, and significant contribution to air pollution levels in the National Capital Region (NCR). Noida Sector 125 is a rapidly developing area characterised by the presence of educational institutions, commercial establishments, and major transportation routes. The region experiences high vehicular traffic, which contributes significantly to the emission of air pollutants such as particulate matter ($PM_{2.5}$ and PM_{10}) and nitrogen dioxide (NO_2). In addition, ongoing construction activities and road dust further increase the concentration of coarse particles (PM_{10}). Due to its proximity to major highways and urban infrastructure, Sector 125 often exhibits fluctuating air quality conditions influenced by traffic intensity and meteorological factors.

Noida Sector 62, on the other hand, is a well-established industrial and commercial hub that houses numerous IT companies, office complexes, and small-scale industries. The area is characterised by continuous vehicular movement and emissions from nearby industrial sources. As a result, it experiences relatively higher levels of particulate matter and gaseous pollutants. $PM_{2.5}$ concentrations are mainly influenced by combustion-related activities, while PM_{10} levels are associated with dust and industrial emissions [53]. Nitrogen dioxide (NO_2) levels are primarily affected by vehicular exhaust and fuel combustion processes.

The air quality data used in this study is obtained from official monitoring stations operated by the Central Pollution Control Board (CPCB) [54] and the State Pollution Control Board (SPCB). These stations provide continuous measurements of key pollutants, including $PM_{2.5}$, PM_{10}, and NO_2. The dataset consists of time-series observations collected over a significant period https://airquality.cpcb.gov.in/ccr/#/login , ensuring sufficient data for analysis and model training. The collected data undergoes preprocessing steps such as handling missing values, normalisation, and outlier detection to improve data quality and reliability. This processed dataset is then used for further analysis and prediction using advanced machine learning and deep learning models. The availability of high-quality and continuous data from these locations makes them suitable for studying air pollution patterns and developing accurate forecasting models [55].

4.2 Exploratory data analysis (EDA)

The present study considers two important locations in Noida: Sector 125 and Sector 62 Fig. 2. These areas represent different types of urban development and are significant for understanding variations in air quality. Noida Sector 125 is primarily an institutional and educational hub, housing several universities, colleges, and corporate offices [56]. The area experiences moderate traffic flow, mainly due to the daily commuting of students and working professionals. The presence of open spaces and relatively planned infrastructure contributes to comparatively better environmental conditions, although pollution levels are still influenced by vehicular emissions and nearby urban activities. On the other hand, Noida Sector 62 is a major commercial and industrial area, known for its IT parks, business centres, and corporate offices. It experiences high traffic density throughout the day due to office commuting and transportation activities. The concentration of commercial establishments and the continuous movement of vehicles contribute significantly to air pollution levels in this region. The datasets collected from these locations include key air pollutants such as PM2.5, PM10, Nitrogen Dioxide (NO_2). These parameters are essential indicators for assessing air quality and understanding environmental conditions in urban regions. The selection of these two locations provides a comparative perspective between an institutional zone and a commercial-industrial area, helping in understanding the impact of different urban activities on air quality.

4.2.1 Statistical exploratory analysis of Noida Sector 125

The statistical summary of air pollutants in Noida Sector 125 and Sector 62 reveals significant variability in pollutant concentrations. Among the pollutants, PM_{10} exhibits the highest mean value (180.94), indicating a strong presence of coarse particulate matter, likely due to dust, construction activities, and vehicular emissions. $PM_{2.5}$ shows

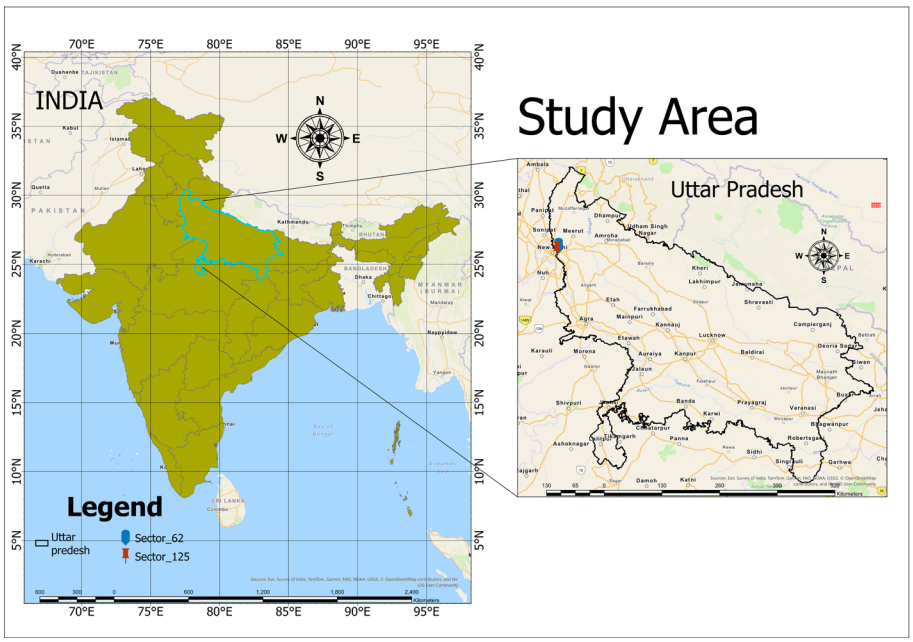


Fig. 2 Study area map showing Delhi and selected locations

Table 1 Statistical summary of air pollutants (Noida Sector 125)

	$PM_{2.5}$	NO_2	PM_{10}
Count	694	694	694
Mean	80.57	49.79	180.94
Std Dev	73.91	36.79	111.67
Min	4.09	1.44	8.71
25%	32.06	20.31	99.04
50% (Median)	49.13	41.35	153.15
75%	101.25	71.55	244.87
Max	407.57	231.58	593.47

Table 2 Statistical summary of air pollutants (Noida Sector 62)

	$PM_{2.5}$	NO_2	PM_{10}
Count	707	707	707
Mean	81.30	16.36	186.26
Std Dev	65.66	11.68	106.85
Min	8.65	0.57	23.17
25%	34.03	8.30	96.51
50% (Median)	60.89	13.19	173.09
75%	107.82	21.11	251.93
Max	464.09	108.50	739.09

a moderate mean concentration (80.57), suggesting the influence of fine particulate sources such as combustion and traffic emissions Table 1. NO_2 has a comparatively lower mean (49.79), but its presence still reflects the impact of vehicular exhaust and industrial activities. The standard deviation values indicate considerable fluctuations in pollutant levels, especially for PM_{10} and $PM_{2.5}$, highlighting inconsistent air quality conditions over time. The wide range between minimum and maximum values further confirms the occurrence of extreme pollution events, with PM_{10} reaching as high as 593.47 and $PM_{2.5}$ up to 407.57. The percentile distribution shows that 50% of $PM_{2.5}$ values lie below 49.13, while PM_{10} median is 153.15, indicating relatively high baseline pollution levels. The 75th percentile values suggest frequent high pollution episodes, particularly for PM_{10} and NO_2. Overall, the data indicate that particulate matter, especially PM_{10}, is the dominant pollutant in the study area, posing significant environmental and health concerns. These variations emphasise the need for continuous monitoring and effective pollution control strategies in urban regions like Noida.

4.2.2 Statistical exploratory analysis of Noida Sector 62

The statistical summary of air pollutants for this dataset highlights notable variations in pollution levels across Noida Sector 125 and Sector 62. PM_{10} shows the highest mean concentration (186.26), indicating a dominant presence of coarse particulate matter, which is typically associated with road dust, construction activities, and mechanical processes Table 2. $PM_{2.5}$ has a mean value of 81.30, suggesting a considerable contribution

from fine particulate sources such as vehicular emissions and fuel combustion. In contrast, NO_2 exhibits a relatively lower mean concentration (16.36), indicating comparatively lesser but still significant impact from vehicular and industrial emissions. The standard deviation values reflect moderate to high variability, particularly in $PM_{2.5}$ and PM_{10} concentrations, suggesting fluctuating air quality conditions over time. The difference between minimum and maximum values is substantial, especially for PM_{10} (23.17 to 739.09) and $PM_{2.5}$ (8.65 to 464.09), indicating the occurrence of extreme pollution episodes. The percentile distribution reveals that 50% of $PM_{2.5}$ values are below 60.89, while PM_{10} has a median value of 173.09, highlighting persistently elevated pollution levels. The 75th percentile values indicate frequent exposure to higher pollution concentrations, especially for particulate matter. NO_2 shows comparatively lower spread but still demonstrates variability across observations. Overall, the analysis suggests that particulate matter, particularly PM_{10}, remains the most critical pollutant in the study area. The presence of high variability and extreme values emphasises the need for effective air quality management strategies and continuous monitoring in rapidly urbanising regions like Noida.

4.2.3 Graphical exploratory analysis of Noida Sector 125

PM2.5 & PM10: Strong positive correlation (0.78). This indicates that both particulate matter pollutants tend to increase and decrease together, likely due to common sources such as vehicular emissions and dust Fig. 3. PM2.5 & NO_2: Moderate positive correlation (0.49). This suggests that traffic and combustion-related activities contribute to both pollutants. PM10 & NO_2: Moderate correlation (0.46), indicating some shared emission sources, particularly from urban activities.

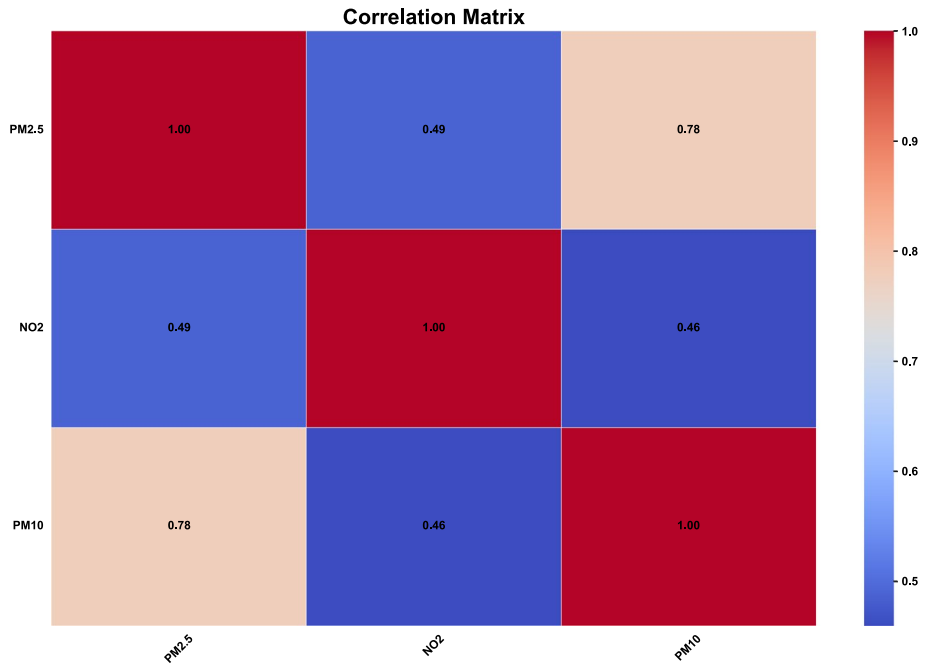


Fig. 3 Distribution of air quality and meteorological variables of Noida Sector 125

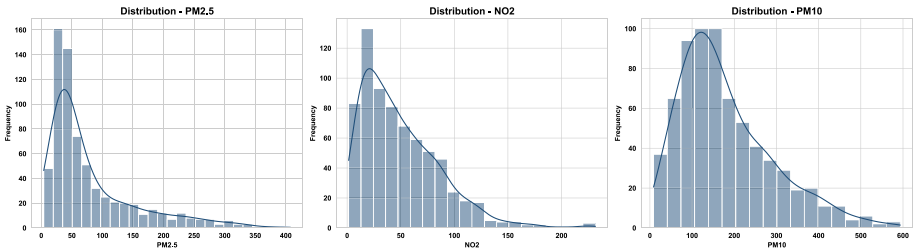


Fig. 4 Distribution of air quality and meteorological variables of Noida Sector 125

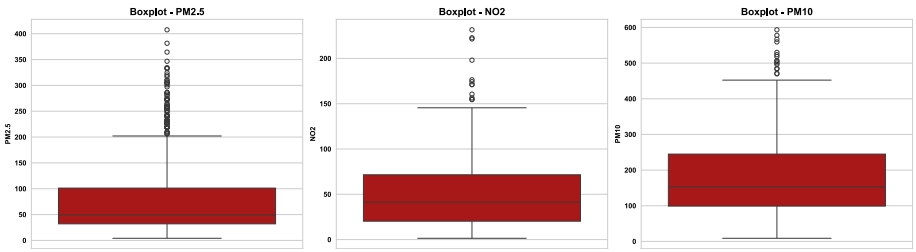


Fig. 5 Distribution of air quality and meteorological variables of Noida Sector 125

PM2.5: Shows a wide distribution with high frequency across different concentration ranges, indicating frequent fluctuations and possible high pollution episodes. NO_2: Exhibits a relatively narrower distribution compared to particulate matter, suggesting more stable concentration levels with fewer extreme values. PM10: Displays a broad distribution similar to PM2.5, indicating high variability and strong presence of coarse particulate matter Fig. 4.

PM2.5: Shows a large interquartile range with several outliers, indicating high variability and extreme pollution events. NO_2: Displays a relatively compact boxplot with fewer outliers, suggesting consistent concentration levels. PM10: Exhibits significant spread and multiple outliers, reflecting high variability similar to PM2.5 Fig. 5. PM2.5 and PM10 are strongly correlated and highly variable, indicating they are major contributors to air pollution. NO_2: Moderately correlated with particulate matter and relatively stable, suggesting localised emission sources such as traffic.

4.2.4 Graphical exploratory analysis of Noida Sector 62

PM2.5 & PM10: Very strong positive correlation (0.92). This indicates that both particulate matter pollutants are highly dependent on common sources such as vehicular emissions, construction activities, and road dust. PM2.5 & NO_2: Moderate positive correlation (0.46). This suggests that combustion-related activities, especially traffic emissions, influence both pollutants. PM10 & NO_2: Moderate correlation (0.44), indicating partial contribution from shared emission sources Fig. 6.

PM2.5: Exhibits a wide distribution with high frequency values, indicating frequent variation and presence of elevated pollution levels. NO_2: Shows a relatively narrower distribution with fewer extreme values, suggesting more stable concentration levels.

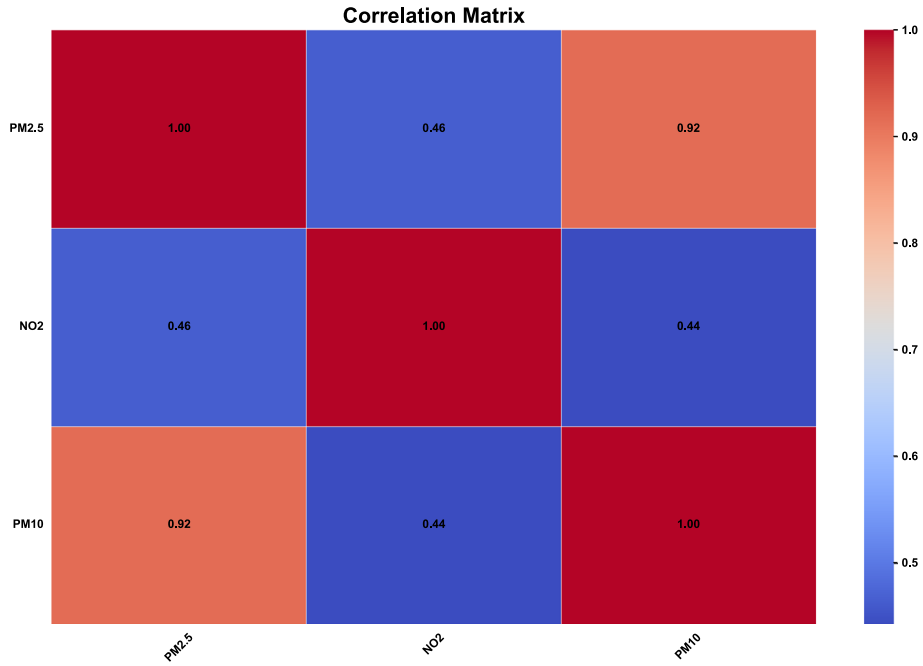


Fig. 6 Distribution of air quality and meteorological variables of Noida Sector 62

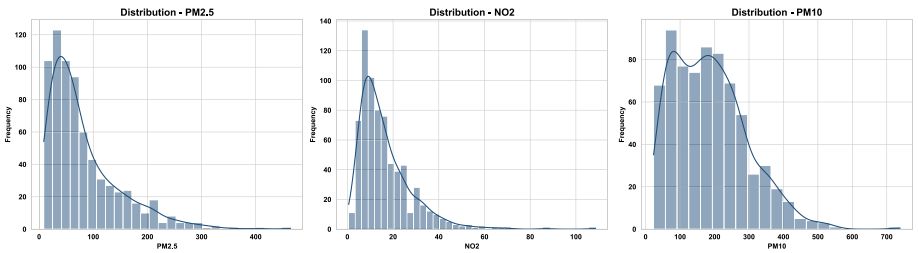


Fig. 7 Distribution of air quality and meteorological variables of Noida Sector 62

PM10: Displays a broad distribution similar to PM2.5, indicating significant variability and frequent high concentration levels Fig. 7.

PM2.5: Shows a large interquartile range with multiple outliers, indicating high variability and extreme pollution events. NO_2: Displays a compact boxplot with fewer outliers, suggesting stable concentration levels. PM10: Exhibits significant spread and numerous outliers, reflecting high variability similar to PM2.5 Fig. 8. PM2.5 and PM10: Strongly correlated and dominant contributors to air pollution with high variability. NO_2: Moderately correlated with particulate matter and relatively stable, indicating localised emission sources such as traffic.

5 Deep learning models

Deep learning models are a subset of machine learning techniques that utilise artificial neural networks with multiple hidden layers to learn complex patterns from large datasets. These models are capable of automatically extracting features and representations, making them highly effective for tasks such as image recognition, natural language processing, and

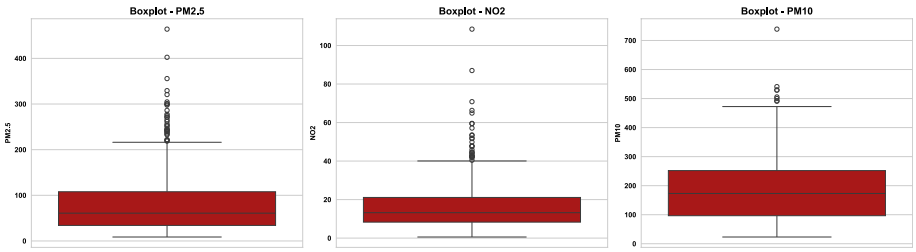


Fig. 8 Distribution of air quality and meteorological variables of Noida Sector 62

time-series forecasting. In air quality analysis, deep learning models can capture nonlinear relationships between pollutants and meteorological factors. Common architectures include Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (Bi-LSTM) and Gated Recurrent Units (GRU). Their ability to handle large-scale data and temporal dependencies makes them suitable for accurate prediction and analysis.

5.1 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are primarily used for processing structured grid data such as images. They use convolutional layers to automatically extract spatial features and patterns. In air quality studies, CNNs can be applied to spatial pollutant distribution and environmental mapping tasks. Their ability to reduce dimensionality while preserving important features makes them highly efficient.

$$y(i,j) = \sum_{m} \sum_{n} x(i-m, j-n) \cdot w(m,n) \tag{1}$$

where: $x(i, j)$ = input feature map $w(m, n)$ = convolution kernel (filter) $y(i, j)$ = output feature map m, n = kernel dimensions

5.2 Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNNs) are designed for sequential and time-series data. They maintain a memory of previous inputs using feedback connections, making them suitable for temporal data analysis. In air quality prediction, RNNs are used to model pollutant trends over time. However, they suffer from vanishing gradient problems when dealing with long sequences.

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \tag{2}$$

$$y_t = W_y h_t \tag{3}$$

where: h_t = hidden state at time t h_{t-1} = previous hidden state x_t = input at time t W_h, W_x, W_y = weight matrices b = bias term y_t = output at time t

5.3 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is an advanced form of RNN that overcomes the limitations of standard RNNs. It uses memory cells and gating mechanisms (input, forget, and

output gates) to retain long-term dependencies. LSTM models are widely used in air quality forecasting because they can capture long-term temporal relationships between pollutants and meteorological parameters.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{4}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{5}$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{6}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \tag{7}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{8}$$

$$h_t = o_t \cdot \tanh(C_t) \tag{9}$$

where: f_t = forget gate i_t = input gate o_t = output gate C_t = cell state $\tilde{C}_t$ = candidate cell state h_t = hidden state x_t = input at time t W_f, W_i, W_o, W_c = weight matrices b_f, b_i, b_o, b_c = bias terms σ = sigmoid activation function

5.4 Gated Recurrent Unit (GRU)

Gated Recurrent Units (GRUs) are a simplified version of LSTM networks. They use fewer gates (reset and update gates), which reduces computational complexity while maintaining performance. GRUs are effective for time-series prediction tasks such as air pollution forecasting and are faster to train compared to LSTM models.

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \tag{10}$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \tag{11}$$

$$\tilde{h}_t = \tanh(W[r_t \cdot h_{t-1}, x_t]) \tag{12}$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \tag{13}$$

where: z_t = update gate r_t = reset gate $\tilde{h}_t$ = candidate hidden state h_t = current hidden state h_{t-1} = previous hidden state x_t = input W_z, W_r, W = weight matrices σ = sigmoid function

5.5 Bidirectional Long Short-Term Memory (Bi-LSTM)

Bidirectional Long Short-Term Memory (Bi-LSTM) extends the LSTM architecture by processing data in both forward and backward directions. This allows the model to capture past and future context simultaneously. In air quality modelling, Bi-LSTM improves prediction accuracy by considering complete temporal dependencies in pollutant data.

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \tag{14}$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \tag{15}$$

$$\tilde{h}_t = \tanh(W[r_t \cdot h_{t-1}, x_t]) \tag{16}$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \tag{17}$$

where: $\overrightarrow{h_t}$ = forward hidden state $\overleftarrow{h_t}$ = backward hidden state x_t = input at time t y_t = output W = weight matrix $[\cdot]$ = concatenation operation

The flowchart presented in Fig. 9 illustrates a systematic framework for air quality prediction using deep learning techniques. The process begins with the collection of pollution and meteorological parameters from reliable sources such as the CPCB. This is followed by EDA, which is divided into numerical and graphical analysis. Numerical EDA includes statistical analysis and handling missing values through imputation, while graphical EDA involves visualisation techniques such as correlation heat maps and bar plots. After preprocessing, a clean dataset is obtained, which is then divided into training, validation, and testing subsets. The training data is used to build deep learning models, and the validation dataset is utilised to evaluate model performance and guide hyperparameter tuning. An iterative learning process is adopted, where the model is repeatedly trained and validated until the validation and training losses stabilise, indicating optimal performance. If the model does not meet performance criteria, hyperparameter tuning is performed to improve its accuracy. Once an effective model is achieved, it is selected and used for predicting the test dataset. Finally, the performance of the model is evaluated using standard metrics such as RMSE, MAE, MAPE, and R2, ensuring a comprehensive assessment of prediction accuracy and model reliability.

6 Methodology

The flowchart illustrated in Fig. 10 outlines a systematic framework for air quality prediction using deep learning techniques. The process begins with the collection of key air pollution parameters, including $PM_{2.5}$, PM_{10}, and NO_2. In the data preprocessing stage, the raw dataset undergoes data cleaning to remove inconsistencies and noise, followed by missing data imputation to handle incomplete values. Subsequently, transformation techniques such as normalisation and scaling are applied to ensure that the data is suitable for model training.

After preprocessing, the dataset is divided into three subsets: training (Y_{Train}), validation, and testing (Y_{Test}) to enable effective model development and unbiased evaluation. In the model learning phase, hybrid deep learning architecture such as combination of bidirectional long short-term memory (Bi-LSTM) and GRU models are employed to capture spatial and temporal patterns in the data. The validation set is used to evaluate model performance during training, and hyperparameter tuning is performed iteratively to improve accuracy. If the model fails to achieve effective performance, the tuning process is repeated until satisfactory results are obtained. Once the model demonstrates stable and efficient performance, it is selected as the final model. The selected model is then used to forecast air quality values for the testing dataset (Y_{Test}), while predictions are also generated for the training dataset (Y_{Train}) for

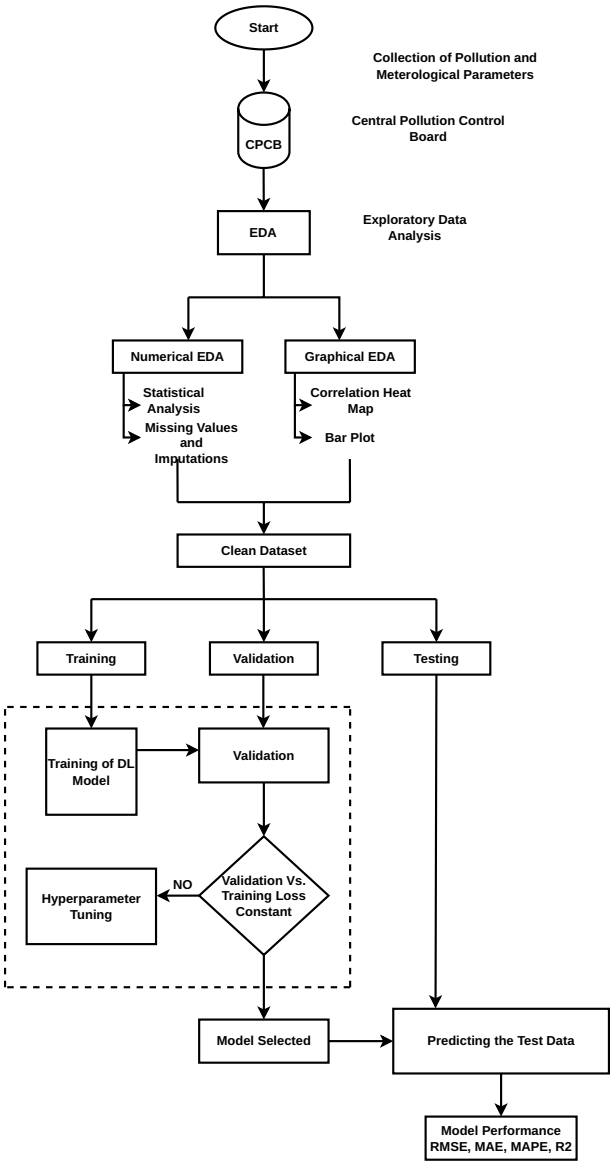


Fig. 9 Overall workflow: data collection, EDA, model training, validation, and testing

comparative analysis. This structured workflow ensures robust model training, optimal parameter selection, and accurate forecasting of air pollution levels.

Problem definition

Let the multivariate time series of air pollutants be defined as:

$$\mathbf{Z}_t = \left[z_t^{(p)}, z_t^{(m)}, z_t^{(n)} \right] \tag{18}$$

where: $z_t^{(p)}$ = PM2.5 concentration, $z_t^{(m)}$ = PM10 concentration and $z_t^{(n)}$ = NO_2 concentration

The prediction objective is:

$$\hat{y}t + \tau = \mathcal{F}(\mathbf{Z}t, \mathbf{Z}t - 1, \ldots, \mathbf{Z}t - \tau) \tag{19}$$

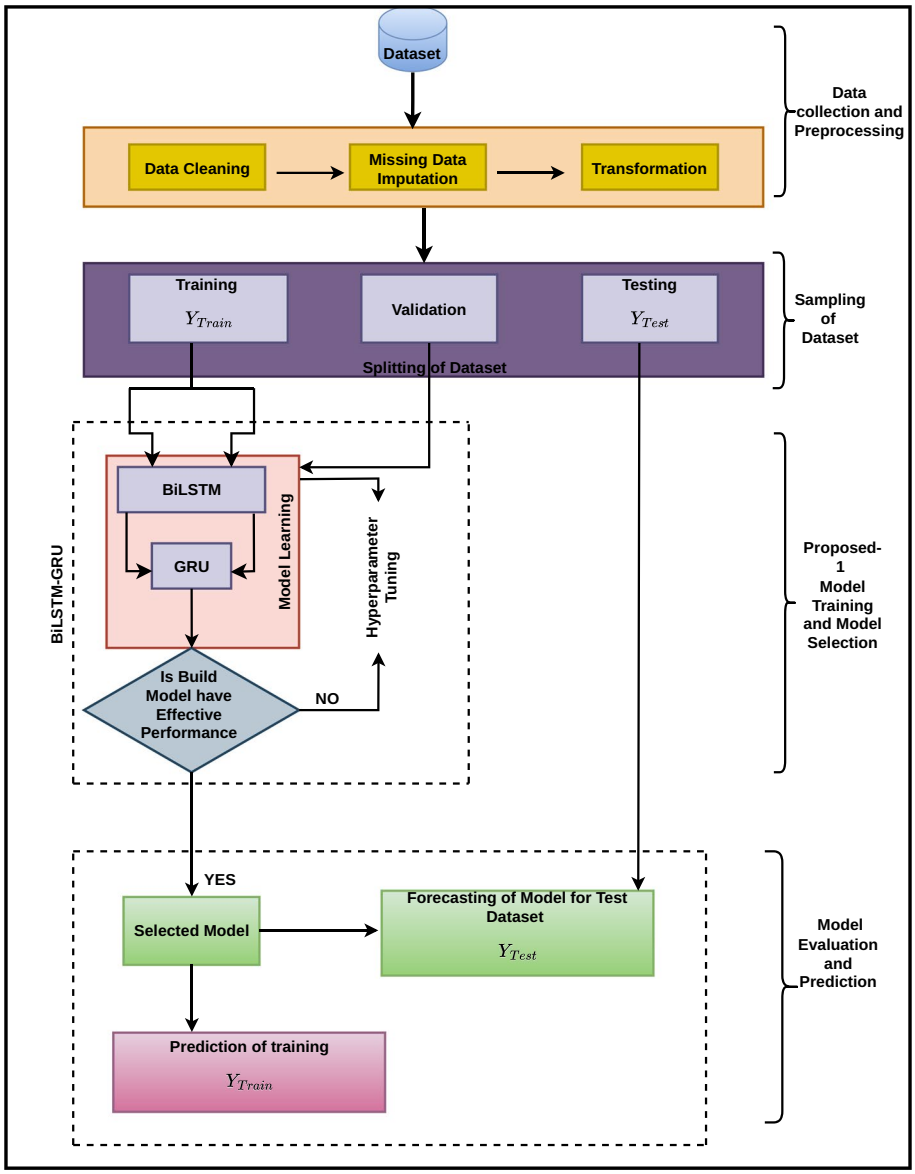


Fig. 10 Proposed framework for air quality prediction using deep learning models

where $\mathcal{F}$ denotes the Hybrid BiLSTM-GRU mapping function.

Data scaling

Each pollutant feature is normalised using Min-Max scaling:

$$\bar{z}_t^{(i)} = \frac{z_t^{(i)} - \alpha^{(i)}}{\beta^{(i)} - \alpha^{(i)}} \tag{20}$$

where $\alpha^{(i)} = \min(z^{(i)})$ and $\beta^{(i)} = \max(z^{(i)})$.

Sequence construction

Input sequence window:

$$\mathbf{S}t = \left[\bar{\mathbf{Z}}t - \tau, \bar{\mathbf{Z}}t - \tau + 1, \ldots, \bar{\mathbf{Z}}t\right] \tag{21}$$

Bidirectional LSTM Layer

Forward Hidden State

$$\overrightarrow{\mathbf{h}}\,t = \mathrm{LSTM}^{\rightarrow}(\overline{\mathbf{Z}}_{\mathrm{t}},\ \overrightarrow{\mathbf{h}}\,\mathrm{t}-1) \tag{22}$$

Backwards Hidden State

$$\overleftarrow{\mathbf{h}}\,t = \mathrm{LSTM}^{\leftarrow}(\overline{\mathbf{Z}}_{\mathrm{t}},\ \overleftarrow{\mathbf{h}}\,\mathrm{t}+1) \tag{23}$$

BiLSTM Output Fusion

$$\mathbf{H}_t = \Phi\left(\overrightarrow{\mathbf{h}}_t,\ \overleftarrow{\mathbf{h}}_t\right) \tag{24}$$

where $\Phi(\cdot)$ denotes concatenation.

Internal LSTM Dynamics

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \cdot \mathbf{u}_t + \mathbf{b}_f) \tag{25}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \cdot \mathbf{u}_t + \mathbf{b}_i) \tag{26}$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \cdot \mathbf{u}_t + \mathbf{b}_c) \tag{27}$$

$$\mathbf{c}t = \mathbf{f}_t \odot \mathbf{c}t - 1 + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \tag{28}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \cdot \mathbf{u}_t + \mathbf{b}_o) \tag{29}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{30}$$

where:

$$\mathbf{u}t = \left[\mathbf{h}t - 1, \bar{\mathbf{Z}}_t\right] \tag{31}$$

GRU Layer

The BiLSTM output is passed to GRU:

$$\mathbf{z}t = \sigma(\mathbf{W}_z\mathbf{H}_t + \mathbf{U}_z\mathbf{g}t - 1 + \mathbf{b}_z) \tag{32}$$

$$\mathbf{r}t = \sigma(\mathbf{W}_r\mathbf{H}_t + \mathbf{U}_r\mathbf{g}t - 1 + \mathbf{b}_r) \tag{33}$$

$$\tilde{\mathbf{g}}t = \tanh(\mathbf{W}_g\mathbf{H}_t + \mathbf{U}_g(\mathbf{r}_t \odot \mathbf{g}t - 1) + \mathbf{b}_g) \tag{34}$$

$$\mathbf{g}t = (1 - \mathbf{z}_t) \odot \mathbf{g}t - 1 + \mathbf{z}_t \odot \tilde{\mathbf{g}}_t \tag{35}$$

Prediction Layer

Final regression output:

$$\hat{y}_{t+1} = \mathbf{W}_y \cdot \mathbf{g}_t + \mathbf{b}_y \tag{36}$$

Loss Function

Mean Squared Error:

$$\mathcal{J} = \frac{1}{M}\sum_{t=1}^{M}\left(y_t - \hat{y}_t\right)^2 \tag{37}$$

Parameter Optimization

Gradient-based update rule:

$$\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} - \lambda\nabla_{\boldsymbol{\theta}}\mathcal{J} \tag{38}$$

Overall Hybrid Model

$$\hat{y}_{t+1} = \Psi(\text{GRU}(\text{BiLSTM}(\mathbf{S}_\text{t}))) \tag{39}$$

Algorithm 1 Hybrid BiLSTM-GRU for air pollution prediction

Require: Multivariate time series data $\mathbf{Z}_t = \{z_t^{(p)}, z_t^{(m)}, z_t^{(n)}\}$
Ensure: Predicted pollution value $\hat{y}_{t+1}$
Input: PM2.5 ($z^{(p)}$), PM10 ($z^{(m)}$), NO_2 ($z^{(n)}$)
Normalize data:
for each feature $i \in \{p, m, n\}$ **do**
  $\bar{z}_t^{(i)} = \frac{z_t^{(i)} - \alpha^{(i)}}{\beta^{(i)} - \alpha^{(i)}}$
end for
Construct sequence window:
$\mathbf{S}t = [\bar{\mathbf{Z}}t - \tau, \ldots, \bar{\mathbf{Z}}_t]$
BiLSTM Forward Pass:
for $t = 1$ to T **do**
  $\overrightarrow{\mathbf{h}}\,t = \text{LSTM}^{\rightarrow}(\bar{\mathbf{Z}}_t, \overrightarrow{\mathbf{h}}\,t - 1)$
end for
BiLSTM Backward Pass:
for $t = T$ to 1 **do**
  $\overleftarrow{\mathbf{h}}\,t = \text{LSTM}^{\leftarrow}(\bar{\mathbf{Z}}_t, \overleftarrow{\mathbf{h}}\,t + 1)$
end for
Concatenate states:
for $t = 1$ to T **do**
  $\mathbf{H}_t = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]$
end for
GRU Layer Processing:
for $t = 1$ to T **do**
  $\mathbf{z}t = \sigma(\mathbf{W}_z\mathbf{H}_t + \mathbf{U}_z\mathbf{g}t - 1)$
  $\mathbf{r}t = \sigma(\mathbf{W}_r\mathbf{H}_t + \mathbf{U}_r\mathbf{g}t - 1)$
  $\tilde{\mathbf{g}}t = \tanh(\mathbf{W}_g\mathbf{H}_t + \mathbf{U}_g(\mathbf{r}_t \odot \mathbf{g}t - 1))$
  $\mathbf{g}t = (1 - \mathbf{z}_t) \odot \mathbf{g}t - 1 + \mathbf{z}_t \odot \tilde{\mathbf{g}}_t$
end for
Prediction Layer:
$\hat{y}_{t+1} = \mathbf{W}_y \cdot \mathbf{g}_T + \mathbf{b}_y$
Loss Computation:
$\mathcal{J} = \frac{1}{M}\sum_{t=1}^{M}(y_t - \hat{y}_t)^2$
Update Parameters:
$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \lambda\nabla_{\boldsymbol{\theta}}\mathcal{J}$
return $\hat{y}_{t+1}$

7 Experimental setup

The experimental setup for multi-pollutant air quality forecasting using deep learning models consists of a computational environment designed to efficiently process large-scale time-series data and train deep learning architectures. The setup includes both hardware and software components that ensure smooth model training, evaluation, and visualisation.

7.1 Hardware and software required

The hardware configuration provides sufficient computational capability to handle large-scale time-series datasets and iterative model training processes. The use of a multi-core CPU and adequate RAM ensures efficient preprocessing and feature engineering, while optional GPU acceleration significantly reduces training time for deep learning models such as LSTM Table 3. On the software side, Python is used as the primary programming language due to its extensive ecosystem for data analysis and machine learning. Libraries such as NumPy and Pandas facilitate efficient data manipulation, while Matplotlib and Seaborn are used for visualisation of temporal trends and model performance. The LSTM model is implemented using TensorFlow/Keras or PyTorch, which provide robust tools for designing and training deep neural networks. Additionally, Scikit-learn is utilised for data preprocessing, scaling, and evaluation metrics. Overall, the integrated hardware and software setup ensures reliable, scalable, and reproducible experimentation for hourly multi-pollutant air quality forecasting.

7.2 Parameter settings of deep learning models

The performance metrics used in this study provide a comprehensive quantitative evaluation of prediction accuracy. RMSE and MAE measure the average magnitude of prediction errors, with RMSE giving higher weight to larger errors Table 4. MAPE expresses the prediction error in percentage form, making it easier to interpret relative error. The coefficient of determination (R^2) evaluates how well the predicted values explain the variance in the observed data, with values closer to 1 indicating better model performance.

8 Results

8.1 Quantitative analysis

The quantitative analysis is performed to evaluate the predictive performance of different deep learning models on the Noida Sector 62 and 125 dataset. This evaluation is primarily based on Mean Absolute Error (MAE), which measures the average magnitude of errors between predicted and actual pollution values. Lower MAE values indicate higher prediction accuracy and better model performance. In this study, the proposed Hybrid BiLSTM-GRU model is compared with traditional models such as BiLSTM, CNN, GRU, LSTM, and RNN across major pollutants, including PM2.5, PM10, and NO_2. The comparison also includes percentage improvement to highlight the effectiveness of the hybrid approach over individual models. The results provide a clear understanding of how well each model captures temporal dependencies and pollutant variations in

Table 3 Hardware and software configuration for experimental setup

Component	Specification
Hardware configuration	
Processor (CPU)	Intel Core i5/i7 or equivalent
RAM	8 GB / 16 GB
Storage	256 GB / 512 GB SSD
GPU (Optional)	NVIDIA GPU (e.g., GTX 1650 / RTX series)
Operating System	Windows 10 / Linux (Ubuntu)
Software configuration	
Programming Language	Python (3.8 or above)
Libraries	NumPy, Pandas, Matplotlib, Seaborn
Deep Learning Framework	TensorFlow / Keras / PyTorch
IDE/Environment	Jupyter Notebook / Google Colab / VS Code
Data Processing Tools	Scikit-learn
Visualization Tools	Matplotlib, Seaborn

Table 4 Performance metrics used for quantitative evaluation of prediction accuracy

Metric	Formula	Description of variables
RMSE	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	y_i = Actual value, $\hat{y}_i$ = Predicted value, n = Number of observations
MAE	$MAE = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	y_i = Actual value, $\hat{y}_i$ = Predicted value, n = Total number of observations
MAPE	$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$	y_i = Actual value, $\hat{y}_i$ = Predicted value, n = Number of observations
R^2	$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$	y_i = Actual value, $\hat{y}_i$ = Predicted value, $\bar{y}$ = Mean of actual values, n = Number of observations

real-world data. This analysis helps in identifying the most reliable model for accurate air quality prediction in urban environments like Noida Sector 62 and 125.

The comparative evaluation presented in Table 5 demonstrates that the proposed Hybrid_BiLSTM_GRU model consistently outperformed the individual deep learning models across most evaluation metrics and pollutant parameters. For PM2.5 prediction, the hybrid model achieved substantial improvements in MSE, RMSE, and MAE compared to CNN, with percentage improvements of 41.03%, 23.21%, and 21.71%, respectively. Similarly, for PM10 forecasting, the hybrid model reduced MSE by 27.93% over CNN and achieved lower RMSE and MAE values than all standalone models. In the case of NO_2, the hybrid architecture also demonstrated competitive performance, although slight negative improvements were observed against RNN and BiLSTM for certain metrics, indicating that these models performed marginally better for specific NO_2 scenarios. Furthermore, the coefficient of determination (R^2) values indicate a significant enhancement in predictive capability, especially for PM2.5, where the hybrid model achieved improvements of 301.98% and 92.47% over CNN and LSTM, respectively. Overall, the Hybrid_BiLSTM_GRU framework exhibited superior robustness, lower prediction errors, and higher explanatory power compared to conventional deep

learning approaches, confirming its effectiveness for multi-pollutant air quality forecasting at Noida Sector 62 as shown in Table 5.

The comparative results presented in Table 6 indicate that the proposed Hybrid_BiLSTM_GRU model achieved superior predictive performance compared to the standalone deep learning models across most pollutants and evaluation metrics at Noida Sector 125. For PM2.5 forecasting, the hybrid model demonstrated substantial reductions in prediction error, achieving improvements of 43.08% in MSE, 24.55% in RMSE, and 24.61% in MAE over the CNN model. Similarly, for PM10 prediction, the hybrid architecture achieved notable improvements, particularly against CNN and LSTM models, with MSE reductions of 28.49% and 13.58%, respectively. In NO_2 prediction, the hybrid model exhibited remarkable enhancement over CNN and LSTM, with MSE improvements reaching 42.58% and 56.52%, respectively, while also producing lower RMSE and MAE values. Furthermore, the R^2 values confirm the strong predictive capability of the hybrid framework, especially for PM10 and NO_2, where the model consistently achieved higher coefficients of determination than the individual models. Although a few negative percentage improvements were observed for PM2.5 and NO_2 R^2 comparisons against CNN and LSTM due to their negative baseline values, the hybrid model still maintained the highest overall prediction accuracy and robustness. Overall, the Hybrid_BiLSTM_GRU model provided more stable and reliable air quality forecasting performance than conventional CNN, RNN, LSTM, GRU, and BiLSTM models, as demonstrated in Table 6.

8.2 Graphical analysis

Graphical analysis is carried out to visually interpret the performance of different deep learning models for air quality prediction across two major locations in Noida, namely Sector 125 and Sector 62. In this study, line plots are used to compare the actual and predicted values of PM2.5 concentrations. The models evaluated include GRU, LSTM, BiLSTM, RNN, and CNN. These graphical representations help in understanding how well each model captures the temporal variations, peak pollution levels, and fluctuations in PM2.5 concentrations. By analysing the closeness between actual and predicted curves, the prediction accuracy and reliability of each model can be effectively assessed. This approach also highlights differences in model performance across the two locations, providing insights into spatial variability in air quality patterns.

8.2.1 Predictive pattern analysis

Predictive pattern analysis is performed to evaluate the capability of deep learning models in capturing temporal variations of air quality across different locations, namely Noida Sector 125 and Noida Sector 62. The analysis focuses on comparing the actual and predicted values of $PM_{2.5}$ concentrations using line plots. The dataset consists of hourly observations collected for recent time periods in 2026. It includes major air pollutants such as $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, and O_3. These variables play a crucial role in determining air quality trends. The graphical analysis using line plots enables visualisation of how well different models capture fluctuations, peaks, and trends in pollutant concentration over time. Models considered in this study include GRU, LSTM, BiLSTM, RNN, and CNN. Overall, recurrent-based models show better performance in capturing

Table 5 Percentage improvement of different deep learning models over the Hybrid_BiLSTM_GRU model for various air quality prediction metrics at Noida Sector 62

Metric	Parameter	CNN	%Imp	RNN	%Imp	LSTM	%Imp	GRU	%Imp	BiLSTM	%Imp	Hybrid_BiLSTM_GRU
MSE	PM2.5	2852.893	41.0396	1874.612	10.2707	2427.252	30.7003	2337.879	28.0511	2024.907	16.9307	1682.077
	PM10	5758.020	27.9367	4373.216	5.1175	4812.027	13.7699	4191.897	1.0134	4524.649	8.2931	4149.416
	NO_2	73.9425	23.0550	55.7263	−2.0973	68.3295	16.7343	73.1573	22.2291	56.2741	−1.1034	56.8951
RMSE	PM2.5	53.4125	23.2143	43.2968	5.2744	49.2671	16.7536	48.3516	15.1773	44.9990	8.8576	41.0131
	PM10	75.8816	15.1099	66.1303	2.5924	69.3688	7.1398	64.7449	0.5080	67.2655	4.2363	64.4160
	NO_2	8.5990	12.2817	7.4650	−1.0432	8.2662	8.7499	8.5532	11.8122	7.5016	−0.5502	7.5429
MAE	PM2.5	39.2946	21.7188	32.4260	5.1369	36.8038	16.4207	36.0987	14.7884	33.9369	9.3602	30.7603
	PM10	58.2105	14.8666	51.2225	3.2523	53.3262	7.0690	49.6178	0.1234	52.1481	4.9697	49.5566
	NO_2	6.2886	13.8939	5.5265	2.0206	6.0021	9.7833	5.9886	9.5798	5.5861	3.0653	5.4149
MAPE	PM2.5	27.4443	15.7354	22.7545	−1.6319	25.4413	9.1011	24.3384	4.9821	24.2605	4.6769	23.1258
	PM10	21.7512	13.9273	19.3781	3.3864	20.3905	8.1834	19.1094	2.0278	19.7706	5.3046	18.7219
	NO_2	35.0277	26.2942	30.3685	14.9860	32.8422	21.3895	28.6224	9.7997	31.7267	18.6255	25.8175
R^2	PM2.5	0.1205	301.9845	0.4221	14.7960	0.2518	92.4771	0.2793	73.4912	0.3758	28.9493	0.4846
	PM10	0.1966	117.6333	0.3898	9.7747	0.3286	30.2257	0.4151	3.0854	0.3687	16.0649	0.4280
	NO_2	0.5290	20.8125	0.6450	−0.9202	0.5648	13.1641	0.5340	19.6810	0.6415	−0.3812	0.6391

Table 6 Percentage improvement of different deep learning models over the Hybrid_BiLSTM_GRU model for various air quality prediction metrics at Noida Sector 125

Metric	Parameter	CNN	%Imp	RNN	%Imp	LSTM	%Imp	GRU	%Imp	BiLSTM	%Imp	Hybrid_BiLSTM_GRU
MSE	PM2.5	7056.3100	43.0874	4569.7420	12.1191	5661.2280	29.0625	4704.5370	14.6370	4436.9940	9.4898	4015.9320
	PM10	11341.7500	28.4924	8370.8350	3.1134	9385.0350	13.5835	9450.9180	14.1859	8213.2020	1.2538	8110.2210
	NO_2	497.5326	42.5801	332.6171	14.1106	657.1348	56.5260	322.7536	11.4858	338.8228	15.6837	285.6829
RMSE	PM2.5	84.0018	24.5595	67.5999	6.2552	75.2411	15.7756	68.5896	7.6079	66.6108	4.8631	63.3714
	PM10	106.4977	15.4378	91.4923	1.5690	96.8764	7.0395	97.2158	7.3641	90.6267	0.6289	90.0568
	NO_2	22.3054	24.2241	18.2378	7.3235	25.6346	34.0652	17.9653	5.9180	18.4071	8.1761	16.9022
MAE	PM2.5	65.2961	24.6172	53.0803	7.2686	58.9129	16.4495	53.1101	7.3207	51.4933	4.4108	49.2221
	PM10	83.5122	16.6596	71.4951	2.6515	75.4953	7.8097	75.1287	7.3597	69.6841	0.1215	69.5994
	NO_2	17.5796	25.2708	14.1766	7.3324	19.5649	32.8536	13.7342	4.3478	13.9056	5.5264	13.1371
MAPE	PM2.5	31.7985	21.6634	26.2883	5.2434	29.8690	16.6028	26.8297	7.1554	26.4980	5.9934	24.9099
	PM10	27.6482	15.8738	23.8625	2.5275	26.4785	12.1575	24.7386	5.9793	24.1750	3.7875	23.2594
	NO_2	29.0997	34.4489	20.2572	5.8351	26.9056	29.1033	20.1284	5.2325	20.3033	6.0488	19.0752
R^2	PM2.5	−0.3879	−155.5590	0.1012	112.9751	−0.1135	−289.8870	0.0747	188.5880	0.1273	69.2923	0.2155
	PM10	0.1150	218.6619	0.3468	5.6698	0.2677	36.9088	0.2626	39.5895	0.3591	2.0506	0.3665
	NO_2	0.1020	365.4452	0.3997	18.8089	−0.1860	−355.2350	0.4175	13.7425	0.3885	22.2345	0.4748

temporal dependencies, while convolution-based models exhibit limitations in handling sequential variations.

The line plot compares actual $PM_{2.5}$ values with predicted outputs from GRU, LSTM, BiLSTM, RNN, and CNN models.

From Fig. 11, it represents a comparative analysis of $PM_{2.5}$ concentration levels for February 2026, showcasing both actual observed values and predictions generated by multiple deep learning models, including RNN, LSTM, GRU, CNN, and BiLSTM.:cont entReference[oaicite:0]index=0 The x-axis represents the timeline of dates throughout the month, while the y-axis indicates $PM_{2.5}$ concentration levels in $\mu g/m^3$. It is evident that all models attempt to capture the temporal variation in pollution levels; however, their performance varies in terms of accuracy and trend alignment. Among the models, BiLSTM and GRU demonstrate relatively closer tracking of the actual $PM_{2.5}$ values, indicating their effectiveness in capturing both short-term and long-term dependencies in time-series data. In contrast, simpler models such as RNN show larger deviations, especially during peaks and fluctuations, suggesting limitations in handling complex temporal patterns. CNN-based predictions exhibit moderate performance, capturing general trends but lacking precision during sharp variations. Overall, the analysis highlights the superiority of advanced recurrent architectures, particularly BiLSTM, for air quality prediction tasks, emphasising their potential role in environmental monitoring and decision-making systems Fig. 12 .

From hyperref [FIG:28]Figure FIG:28, it illustrates a comparative evaluation of $PM_{2.5}$ concentration levels for February 2026, highlighting the actual observed values alongside predictions from various deep learning models, including RNN, LSTM, GRU, CNN, and BiLSTM.:contentReference[oaicite:0]index=0 The x-axis represents the sequence of dates, while the y-axis shows $PM_{2.5}$ concentration levels in $\mu g/m^3$. Compared to the previous analysis, the concentration range in this figure appears slightly lower, indicating moderate pollution levels during the observed period. All models generally follow the trend of actual $PM_{2.5}$ values; however, differences in prediction accuracy are evident. BiLSTM again demonstrates the closest alignment with actual data, effectively capturing both rising and falling trends. GRU and LSTM also perform well, though minor deviations are observed during sudden fluctuations. CNN captures the overall trend but tends to smooth out peaks and valleys, leading to less precise predictions. RNN exhibits the largest deviations, particularly during periods of rapid change, reflecting its limitations in modelling complex temporal dependencies. Overall, the results reinforce that advanced recurrent models, especially BiLSTM, provide more reliable predictions for air quality forecasting.

8.2.2 Prediction distributive analysis

The prediction distribution analysis is performed using superimposed scatter plots of predicted versus actual $PM_{2.5}$ concentrations for different deep learning models at both Noida Sector 125 and Sector 62. These plots provide a visual assessment of model accuracy, dispersion, and correlation by comparing the alignment of data points along the 45-degree reference line. For Noida Sector 125, the scatter plots indicate that the RNN and GRU models exhibit relatively strong linear alignment with the ideal diagonal line,

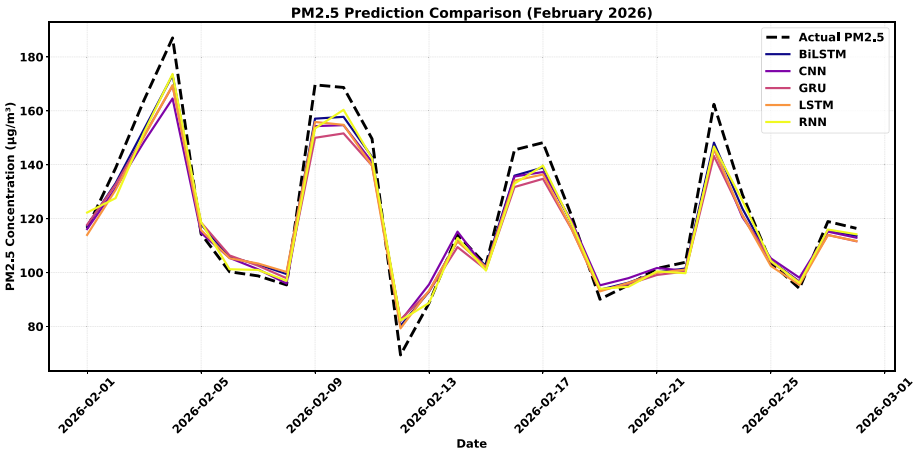


Fig. 11 $PM_{2.5}$ Prediction Comparison using Deep Learning Models (Noida Sector 125)

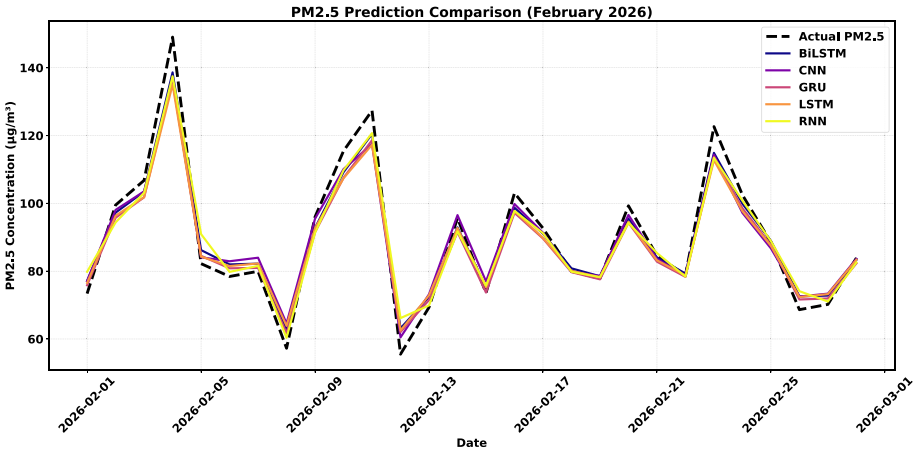


Fig. 12 $PM_{2.5}$ Prediction Comparison using Deep Learning Models (Noida Sector 62)

suggesting better predictive capability and higher correlation. The RNN model demonstrates the most compact clustering of points around the reference line, indicating lower prediction error and strong generalisation ability. The GRU model follows closely, although with slightly higher dispersion at peak concentration levels. In contrast, the BiLSTM and LSTM models show moderate spread, reflecting increased variability in predictions and reduced consistency in capturing extreme $PM_{2.5}$ values. The CNN model displays the highest dispersion and noticeable deviation from the diagonal line, particularly at higher concentrations, indicating its limited effectiveness in modelling temporal dependencies in highly dynamic air quality data. For Noida Sector 62, the scatter plots reveal comparatively improved prediction consistency across most models, as indicated by tighter clustering and better alignment with the reference line. The GRU model exhibits the strongest correlation with the actual values, showing a highly concentrated distribution of points and minimal deviation, confirming its robustness and reliability. The CNN model also demonstrates competitive performance with relatively good alignment, although slight dispersion is observed during peak pollution conditions.

The BiLSTM and RNN models show moderate clustering, capturing the general trend but with noticeable deviations in higher concentration ranges. The LSTM model, while improved compared to Sector 125, still exhibits some spread and lower consistency.

From Fig. 13The scatter plot analysis for $PM_{2.5}$ prediction in Noida Sector 162demonstrates that all deep learning models, including BiLSTM, CNN, GRU, LSTM, and RNN, achieve extremely high prediction accuracy. The data points are closely aligned along the 45-degree diagonal line, indicating a strong agreement between actual and predicted values. BiLSTM and RNN models exhibit a perfect correlation coefficient of 1.00, reflecting near-perfect prediction capability, while CNN, GRU, and LSTM also show very high correlations of approximately 0.99. The tight clustering of points with minimal dispersion suggests very low prediction error and high consistency across varying pollution levels. Overall, the results confirm that all models are highly reliable for $PM_{2.5}$ forecasting in this region, with BiLSTM and RNN demonstrating slightly superior stability and precision.

From Fig. 14. The scatter plot analysis for $PM_{2.5}$ prediction in Noida Sector 62 demonstrates that all deep learning models, including BiLSTM, CNN, GRU, LSTM, and RNN, exhibit strong predictive performance. The plotted points are closely aligned along the 45-degree diagonal line, indicating a high level of agreement between actual and predicted values. Furthermore, the correlation coefficient for all models is approximately 0.99, suggesting a very strong positive relationship and high model accuracy. Minimal dispersion of data points around the diagonal line indicates low prediction error and high consistency across different pollution levels. Among the models, BiLSTM and RNN show slightly tighter clustering, implying relatively better stability, whereas CNN, GRU, and LSTM also perform comparably with only minor deviations. Overall, the scatter plot confirms that the proposed models are highly effective for $PM_{2.5}$ forecasting in the selected region.

8.3 Statistical analysis of results based on measured values and prediction

The Non-Parametric Friedman ranking test can be used for comparing several models and their predictive performance, thus identifying those differences between them that are statistically significant, specifically in relation to the GRU, LSTM, BiLSTM, RNN, and CNN model types. Consequently, it helps to find out which model can prove to be the most efficient under the varied environmental conditions. In order to estimate the model complexity and prevent overfitting, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) can be implemented in analysing its performance. They both introduce penalties in case there are too many parameters in the model, but still value its goodness-of-fit to a particular dataset in the process of estimation. The lower the AIC and BIC indicators are, the more suitable the forecast model of the air quality in question will be. The Diebold–Mariano (DM) test can help to identify any possible differences in the model performance through comparison of their forecast accuracy and precision. Thus, it will be possible to estimate whether the differences between model forecast performance are significant or random at all, which is important for the validation of the results obtained in the course of the research. Also, to provide more detailed information on model performance, the Taylor diagram can be used. As is

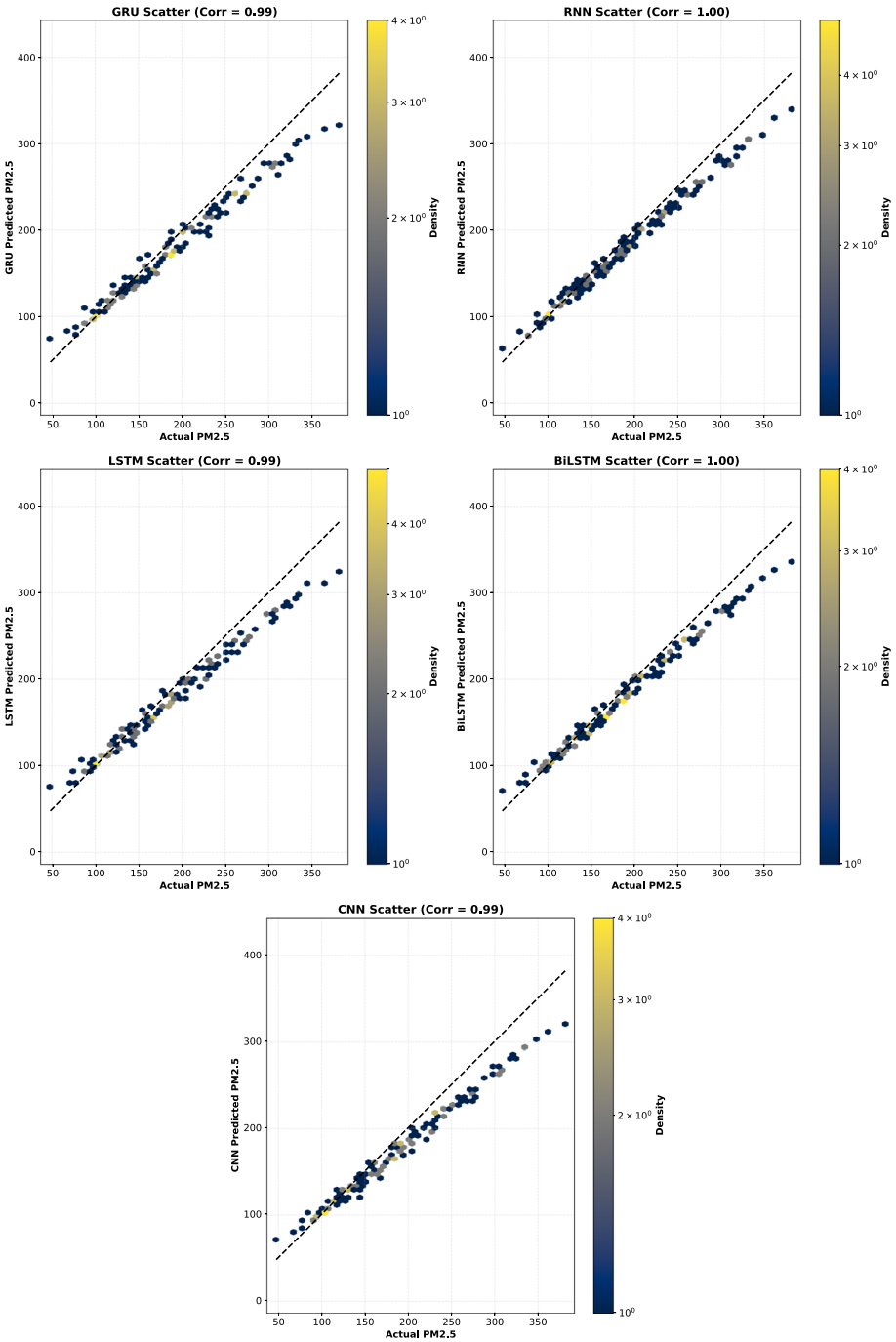


Fig. 13 Superimposed Scatter Plots of Predicted vs Actual $PM_{2.5}$ (Noida Sector 125)

known, it incorporates several basic statistics, including correlation coefficient, standard deviation, and root mean square error (RMSE), into one chart.

8.3.1 AIC-BIC test analysis

The model comparison based on Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), as presented in Noida Sector 125 and Noida Sector 62,

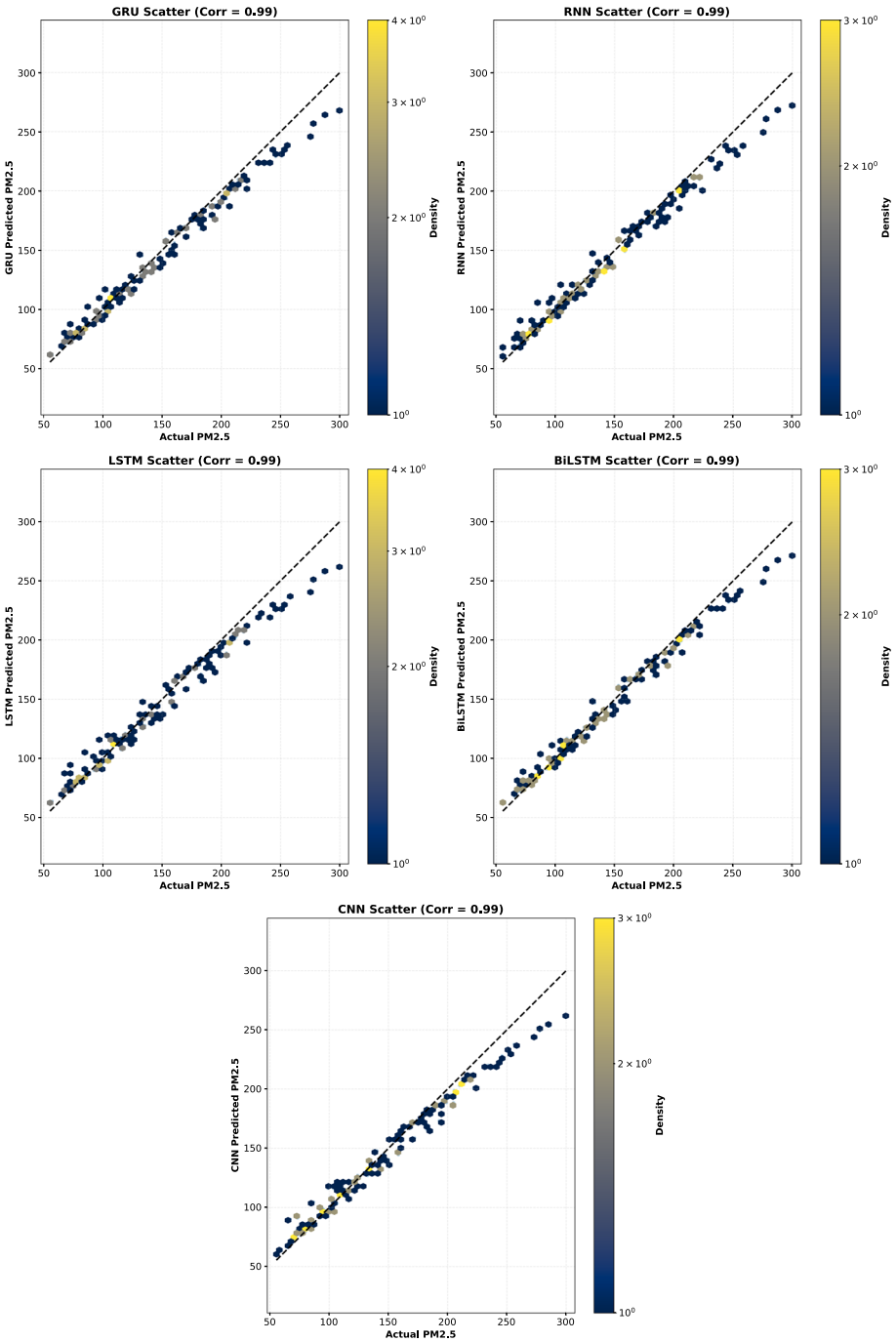


Fig. 14 Superimposed Scatter Plots of Predicted vs Actual $PM_{2.5}$ (Noida Sector 62)

provides insight into the relative goodness-of-fit and model complexity for different deep learning models used in $PM_{2.5}$ prediction. Both AIC and BIC penalise model complexity while rewarding better fit, and lower values indicate superior model performance. From Table 7 of Noida Sector 125, it is observed that the RNN model achieves the lowest AIC and BIC values, suggesting a better trade-off between model fit and complexity for that dataset, followed by BiLSTM. However, in Table 8 of Noida Sector 62, BiLSTM clearly

outperforms all other models by achieving the lowest AIC and BIC values, indicating its superior performance and efficiency in capturing the underlying data patterns. Across both datasets, GRU and LSTM exhibit moderate performance with relatively higher AIC and BIC values, while CNN consistently shows the highest values, indicating poorer model fit and higher complexity. Although RNN performs well in the first dataset, its performance is slightly inferior to BiLSTM in the second dataset. Overall, considering both locations, BiLSTM demonstrates more consistent and robust performance across different scenarios, making it a more reliable model for $PM_{2.5}$ prediction compared to other models.

8.3.2 Non-parametric statistical friedman ranking

The 3D Friedman ranking analysis for Noida Sector 125 in Fig. 15a highlights the comparative performance of deep learning models across multiple evaluation metrics, including R^2, MAPE, MAE, and RMSE. The results indicate that the CNN model consistently achieves the highest rank across all metrics, suggesting superior overall performance. GRU and LSTM follow closely with competitive rankings, demonstrating stable and reliable prediction capabilities. In contrast, RNN obtains the lowest ranks, indicating relatively weaker performance compared to other models. BiLSTM shows a moderate ranking, performing better than RNN but not surpassing CNN, GRU, or LSTM. Overall, the ranking pattern suggests that CNN is the most effective model for $PM_{2.5}$ prediction in Sector 125, while RNN is the least effective. Similarly, the Friedman ranking for Noida Sector 62 in Fig. 15b reveals a slightly different performance distribution among the models. In this case, BiLSTM achieves the best ranking across most evaluation metrics, indicating its strong predictive capability in this region. CNN and GRU also perform well, maintaining competitive rankings, while LSTM shows moderate performance. RNN again ranks lowest, reflecting consistent underperformance across both locations. This variation between Sector 125 and Sector 62 suggests that model effectiveness may depend on regional data characteristics. Overall, BiLSTM emerges as the most reliable model for Sector 62, whereas CNN dominates in Sector 125, highlighting the importance of location-specific model selection Table 10 .

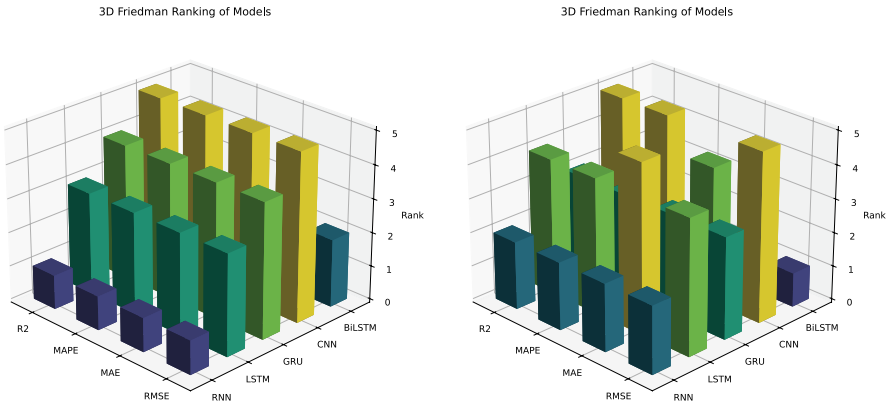
The performance evaluation of different deep learning models based on error metrics and goodness-of-fit, as presented in Noida Sector 125 and Noida Sector 62, provides a comprehensive comparison of their predictive capabilities for $PM_{2.5}$ concentration. The models are assessed using RMSE, MAE, MAPE, and R^2, along with their corresponding rankings. From Table 9, it is observed that the RNN model achieves the best

Table 7 Model comparison using AIC and BIC values (Noida Sector 125)

Model	AIC	BIC
RNN	366.8134	368.7254
BiLSTM	373.2934	375.2055
LSTM	394.5293	396.4413
GRU	395.4069	397.3189
CNN	403.0657	404.9777

Table 8 Model comparison using AIC and BIC values (Noida Sector 62)

Model	AIC	BIC
BiLSTM	330.3994	332.3114
RNN	332.1870	334.0990
GRU	336.9464	338.8584
LSTM	359.1363	361.0483
CNN	363.5587	365.4707



(a) 3D Friedman Ranking of Deep Learning Models for Noida Sector 125

(b) 3D Friedman Ranking of Deep Learning Models for Noida Sector 62

Fig. 15 **a** 3D Friedman Ranking of Deep Learning Models for Noida Sector 125. **b** 3D Friedman Ranking of Deep Learning Models for Noida Sector 62

performance across all metrics, attaining the lowest error values and the highest R^2, thereby securing the top rank. BiLSTM closely follows as the second-best model, while LSTM, GRU, and CNN show progressively poorer performance, with CNN ranking last. In contrast, the results in hyperref [TAB:20]Table TAB:20 indicate that BiLSTM outperforms all other models by achieving the lowest RMSE, MAE, and MAPE values along with the highest R^2, thereby securing the top rank across all evaluation criteria. RNN performs as the second-best model, followed by GRU, while LSTM and CNN exhibit relatively higher error values and lower predictive accuracy. Overall, a comparative analysis of both locations reveals that although RNN performs best in the first dataset, BiLSTM demonstrates more consistent and robust performance across both datasets. This highlights the superior ability of BiLSTM in capturing complex temporal dependencies in air quality data, making it a more reliable model for $PM_{2.5}$ prediction compared to other deep learning approaches.

9 Limitations and future research

Although the proposed Hybrid_BiLSTM_GRU model demonstrated promising performance for multi-pollutant air quality forecasting, several limitations remain in the current study. First, the developed models primarily rely on historical air quality and meteorological datasets collected from CPCB and SPCB monitoring stations. Important

Table 9 Performance metrics and ranking of models

Model	RMSE	MAE	MAPE	R^2	RMSE rank	MAE rank	MAPE rank	R^2 Rank	Avg rank
RNN	10.74497	8.11098	5.39905	0.96134	1	1	1	1	1
BiLSTM	10.79897	8.14973	5.90621	0.96095	2	2	2	2	2
LSTM	13.69983	10.40064	7.36114	0.93715	3	3	3	3	3
GRU	14.11397	10.72232	7.52397	0.93329	4	4	4	4	4
CNN	15.05963	11.20843	7.59456	0.92405	5	5	5	5	5

Table 10 Performance metrics and ranking of models (Dataset 2)

Model	RMSE	MAE	MAPE	R^2	RMSE rank	MAE rank	MAPE rank	R^2 Rank	Avg rank
BiLSTM	6.526	4.966	4.470	0.984	1	1	1	1	1
RNN	6.777	5.125	4.533	0.983	2	2	2	2	2
GRU	7.332	5.435	4.557	0.980	3	3	3	3	3
LSTM	8.822	6.519	5.607	0.971	4	5	4	4	4.25
CNN	9.051	6.419	5.678	0.969	5	4	5	5	4.75

external factors such as real-time traffic density, industrial emission records, construction activities, and satellite-based atmospheric observations were not incorporated, which may influence prediction accuracy. Second, the study was limited to two monitoring locations, Noida Sector 62 and Noida Sector 125, and therefore the generalisability of the model to other geographical regions or climatic conditions requires further validation. In addition, deep learning models generally operate as black-box systems, making the interpretation of learned patterns and decision-making processes difficult. Another limitation is the computational complexity associated with training recurrent and hybrid deep learning architectures, especially for long-term forecasting applications.

Future research can focus on integrating multi-source heterogeneous datasets, including traffic flow information, satellite imagery, IoT sensor networks, and industrial emission data, to improve forecasting robustness and accuracy. Advanced architectures such as Transformer networks, attention-based deep learning models, and explainable artificial intelligence (XAI) techniques can also be explored to enhance model interpretability and transparency. Furthermore, hybrid optimisation methods and ensemble learning approaches may help improve prediction stability and reduce computational overhead. Extending the framework to real-time forecasting systems and deploying it within smart city infrastructures can further support environmental monitoring, early warning systems, and pollution control strategies. Future studies may also investigate transfer learning and federated learning approaches for building scalable and location-independent air quality prediction systems.

10 Conclusion

This study presented a deep learning-based framework for multi-pollutant air quality prediction in Noida Sector 62 and Noida Sector 125 using pollutants such as $PM_{2.5}$, PM_{10}, and NO_2. Several deep learning models, including CNN, RNN, LSTM, GRU, BiLSTM, and the proposed Hybrid_BiLSTM_GRU model, were implemented and evaluated

using MAE, RMSE, MAPE, and R^2 metrics. Experimental results demonstrated that recurrent architectures, particularly GRU and BiLSTM, achieved better forecasting performance compared to conventional deep learning models due to their capability to capture long-term temporal dependencies. Among all models, the proposed Hybrid_BiLSTM_GRU model achieved the best overall performance. At Noida Sector 62, the hybrid model obtained RMSE values of 41.01, 64.41, and 7.54 for $PM_{2.5}$, PM_{10}, and NO_2, respectively, with corresponding R^2 values of 0.4846, 0.4280, and 0.6391. Similarly, at Noida Sector 125, the model achieved RMSE values of 63.37, 90.05, and 16.90, along with R^2 scores of 0.2155, 0.3665, and 0.4748 for $PM_{2.5}$, PM_{10}, and NO_2, respectively. The hybrid framework also achieved up to 43.08% improvement in MSE and 24.61% improvement in MAE compared to conventional models. Overall, the study confirms that hybrid deep learning architectures provide an efficient and reliable solution for urban air quality forecasting and can support intelligent environmental monitoring and pollution management systems.

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Author contributions
Kamred Udham Singh: Visualisation, Writing - Review, Investigation, Editing, Supervision. Abhishek Kushwaha: Conceptualisation of this study, Methodology, Writing - Methodology, Software, Data curation, Writing - Original draft preparation, Visualisation, Writing - Review. Naushad Ahmad: Validation, Formal analysis, Methodology, Investigation, Editing, Supervision.

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Data availability
The datasets (https://airquality.cpcb.gov.in/ccr/#/login) and code generated or analysed during the current study are available from the corresponding author on reasonable request. Availability of GIS data: Download data by country for ArcGIS plot (https://www.diva-gis.org/).

Code availability
All code was implemented in Python. Interested researchers can request access by contacting the corresponding author.

Declarations

Ethics approval and consent to participate
Not applicable. https://diva-gis.org/data.html

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Ghosh B, De M, Padhy PK. India's air pollution burden (2019–2023): Spatial-temporal trends, emissions, weather effects, and adult health risks across state and city. Water Air Soil Pollut. 2026;237(12):701. https://doi.org/10.1007/s11270-026-09368-6.
2. Saxena V. Water quality, air pollution, and climate change: investigating the environmental impacts of industrialization and urbanization. Water Air Soil Pollut. 2025;236(2):73. https://doi.org/10.1007/s11270-024-07702-4.
3. Kumari S, Choudhury A, Karki P, Simon M, Chowdhry J, Nandra A, et al. Next-generation air quality management: unveiling advanced techniques for monitoring and controlling pollution. Aerosol Sci Eng. 2025. https://doi.org/10.1007/s41810-024-00281-1.
4. Aljafen BN, Shaikh N, AlKhalifah JM, Meo SA. Effect of environmental pollutants particulate matter (pm2.5, pm10), nitrogen dioxide (no2), sulfur dioxide (so2), carbon monoxide (co) and ground level ozone (o3) on epilepsy. BMC Neurol, 2025;25(1):133. https://doi.org/10.1186/s12883-025-04142-3.

5. Ishihara Y, Tanaka M, Nezu N, Ishihara N, Oguro A, Vogel CFA. Pathways to the brain: Impact of fine particulate matter components on the central nervous system. Antioxidants. 2025;14(6):730. https://doi.org/10.3390/antiox14060730.
6. Kalantari E, Gholami H, Malakooti H, Nafarzadegan AR, Moosavi V. Machine learning for air quality index (AGI) forecasting: shallow learning or deep learning? Environ Sci Pollut Res. 2024;31(54):62962–82. https://doi.org/10.1007/s11356-024-35404-1.
7. Kohila C, Jeyanthi KMA, Rengan PK. Predicting air quality using a hybrid deep learning model to achieve environmental sustainability. Water Air Soil Pollut. 2026;237(5):267. https://doi.org/10.1007/s11270-025-08951-7.
8. Kumar V, Ahmad N. Deep learning for air quality prediction after covid-19 pandemic based on pollutant and meteorological data. In: 4th International Conference on Communication & Information Processing (ICCIP), 2022. https://doi.org/10.2139/ssrn.4292346.
9. Thangavel P, Park D, Lee YC. Recent insights into particulate matter (pm2.5)-mediated toxicity in humans: an overview. Int J Environ Res Public Health, 2022;19(12):7511. https://doi.org/10.3390/ijerph19127511.
10. Wang X, Singh RP, Eiche E, Jiang F, Pippal PS, Kumar R, et al. Cross-regional characteristics, chemical composition, and source contributions of atmospheric particulate matter in germany and India. Sci Total Environ. 2026;1014:181386. https://doi.org/10.1016/j.scitotenv.2026.181386.
11. Sohail MA, Ahmad N. Machine learning empowered iot devices, analysis of indoor and outdoor temperature and health risks. In: IoT Sensors, ML, AI and XAI: empowering a smarter world, 2024;35–51. Springer. https://doi.org/10.1007/978-3-031-68602-3_3.
12. Mezoue CA, Ngangmo YC, Choudhary A, Monkam D. Measurement of fine particle concentrations and estimation of air quality index (AGI) over northeast Douala, Cameroon. Environ Monitor Assess. 2023;195(8):965. https://doi.org/10.1007/s10661-023-11582-2.
13. de Souza A, de Oliveira-Júnior JF, Cardoso KRA, Gautam S. Impact of vehicular emissions on ozone levels: A comprehensive study of nitric oxide and ozone interactions in urban areas. Geosyst Geoenviron. 2025;4(1):100348. https://doi.org/10.1016/j.geogeo.2024.100348.
14. Kumar V, Chand F, Ahmad N. A novel ensemble loss function of few-shot learning for tomato leaf disease detection. In: 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS), 2024b;1:971–976. IEEE. https://doi.org/10.1109/ICACCS60874.2024.10717060.
15. Shakrullah K, Tariq S, Shirazi SA, Minallah MN, Shahzad H, Mariam A. Spatio-temporal changes of particulate matter (pm 2.5) over brazil and its correlation with meteorological variables. Geograp Environ Sustain, 2025;18(2):82–90. https://doi.org/10.24057/2071-9388-2025-3600.
16. Sarkar N, Gupta R, Keserwani PK, Govil MC. Air quality index prediction using an effective hybrid deep learning model. Environ Pollut. 2022;315:120404. https://doi.org/10.1016/j.envpol.2022.120404.
17. Masood A, Ahmad K. Data-driven predictive modeling of pm2. 5 concentrations using machine learning and deep learning techniques: a case study of delhi, india. Environ Monitor Assess. 2023;195(1):60. https://doi.org/10.1007/s10661-022-10603-w.
18. Hubbell BJ, Kaufman A, Rivers L, Schulte K, Hagler G, Clougherty J, et al. Understanding social and behavioral drivers and impacts of air quality sensor use. Sci Total Environ. 2018;621:886–94. https://doi.org/10.1016/j.scitotenv.2017.11.275.
19. Zeeshan S, Malik MAI. Ai-driven analysis of meteorological and emission characteristics influencing urban smog. Gases. 2026;6(1):10. https://doi.org/10.3390/gases6010010.
20. Pan L, Zhang D, Zhang B, Yuan M, Xu Y. Seasonal pollution characteristics and environmental impact of VOCS in urban areas. Atmos Pollut Res. 2026. https://doi.org/10.1016/j.apr.2026.102990.
21. Chen PJ, Ngo TH, Hsu YC. Integrated analysis of pm2.5 characteristics and multi-pollutant health risks. Sci Total Environ. 2026;1020:181584. https://doi.org/10.1016/j.scitotenv.2026.181584.
22. Ahmad N, Kumar V. Enhancing pm 2.5 air pollution forecasting with novel random imputation based on hybrid rnn-bidirectional gru (nri rnn-bigru) model. SN Comput Sci. 2025b;6(6):637. https://doi.org/10.1007/s42979-025-04167-y.
23. Bernacki J, Scherer R. A comprehensive review of data-driven techniques for air pollution forecasting. Sensors. 2025;25(19):6044. https://doi.org/10.3390/s25196044.
24. Ahmad N, Jas S. Quantum-inspired neural networks for time-series air pollution prediction and control of the most polluted region in the world. Quant Mach Intell. 2025;7(1):9. https://doi.org/10.1007/s42484-024-00232-6.
25. Janiesch C, Zschech P, Heinrich K. Machine learning and deep learning. Electron Mark. 2021;31(3):685–95. https://doi.org/10.1007/s12525-021-00475-2.
26. Mishra RK, Reddy GS. Understanding deep learning: a comprehensive review. Math Probl Eng. 2021;2021:5548884. https://doi.org/10.1155/2021/5548884.
27. Son S, Seok J. Optimal activation function selection for enhanced model performance. Inf Sci. 2026. https://doi.org/10.1016/j.ins.2026.123300.
28. Kumar S, Kour V, Raj A, Tapung T, Mishra S, Misra R, Singh T. Optimizing air pollution forecasting models through knowledge distillation: a novel GCN and trans_gru methodology for indian cities. IEEE Access, 2025b;13:40237–40257. https://doi.org/10.1109/ACCESS.2025.3546504.
29. Khoei TT, Slimane H. Deep learning: systematic review, models, challenges. Neural Comput Appl. 2023;35(31):23103–24. https://doi.org/10.1007/s00521-023-08957-4.
30. Kumar S, Kour V, Verma A, Sah RK, Misra R. Enhanced forecasting accuracy: leveraging gru+ bilstm with attention for precise floodwater level prediction. In 2024 5th International Conference on Image Processing and Capsule Networks (ICIPCN), 2024a;108–115. IEEE. https://doi.org/10.1109/ICIPCN63822.2024.00026.
31. Kumar S, Kour V, Kumar P, Deshmane A, Mishra S, Misra R. Gru-lstm with attention-based forecasting for enhanced air quality. Nat Hazards. 2025;121(13):15925–47. https://doi.org/10.1007/s11069-025-07408-8.
32. Ahmad N, Kumar V. Effective air pollution prediction using Wavenet deep learning with Xgboost (1dcnn-bilstm-xgrc) for urban us embassies. Theoret Appl Climatol. 2025;156(9):464. https://doi.org/10.1007/s00704-025-05715-5.

33. Ofremu GO, Raimi BY, Yusuf SO, Dziwornu BA, Nnabuife SG, Eze AM, et al. Exploring the relationship between climate change, air pollutants and human health: impacts, adaptation, and mitigation strategies. Green Energy Resourc. 2025;3(2):100074. https://doi.org/10.1016/j.gerr.2024.100074.
34. Singh V, Srivastava R, Bhatt AK. Sources of air pollution. In: Battling air and water pollution: protecting our planet's vital resources, 2025;17–29. Springer. https://doi.org/10.1007/978-981-96-4375-2_2.
35. Khajeamiri Y, Sharifi S, Moradpour N, Khajeamiri A. A review on the effect of air pollution and exposure to pm, no2, o3, so2, co and heavy metals on viral respiratory infections. J Air Pollut Health, 2021. https://doi.org/10.18502/japh.v5i4.6445
36. Hayhurst AN, Lawrence AD. Emissions of nitrous oxide from combustion sources. Prog Energy Combust Sci. 1992;18(6):529–52. https://doi.org/10.1016/0360-1285(92)90038-3.
37. Manisalidis I, Stavropoulou E, Stavropoulos A, Bezirtzoglou E. Environmental and health impacts of air pollution: a review. Front Public Health. 2020;8:14. https://doi.org/10.3389/fpubh.2020.00014.
38. Mallet V, Sportisse B. Air quality modeling: From deterministic to stochastic approaches. Comput Math Appl. 2008;55(10):2329–37. https://doi.org/10.1016/j.camwa.2007.11.004.
39. Pielke RA, Avissar R, Raupach M, Dolman AJ, Zeng X, Denning AS. Interactions between the atmosphere and terrestrial ecosystems: influence on weather and climate. Glob Change Biol. 1998;4(5):461–75. https://doi.org/10.1046/j.1365-2486.1998.t01-1-00176.x.
40. Liao Q, Zhu M, Lin W, Pan X, Tang X, Wang Z. Deep learning for air quality forecasts: a review. Curr Pollut Rep. 2020;6(4):399–409. https://doi.org/10.1007/s40726-020-00159-z.
41. Rybarczyk Y, Zalakeviciute R. Machine learning approaches for outdoor air quality modelling: a systematic review. Appl Sci. 2018;8(12):2570. https://doi.org/10.3390/app8122570.
42. Shiri FM, Perumal T, Mustapha N, Mohamed R. A comprehensive overview and comparative analysis on deep learning models: Cnn, rnn, lstm, gru. arXiv preprint arXiv:2305.17473, 2023. https://doi.org/10.48550/arXiv.2305.17473.
43. Sachar DPS. Time series forecasting using deep learning: a comparative study of LSTM, GRU, and transformer models. J Comput Sci Technol Stud. 2023;5(1):74–89. https://doi.org/10.32996/jcsts. 2023.5.1.9
44. Ahmed SF, Alam MSB, Hassan M, Rozbu MR, Ishtiak T, Rafa N, et al. Deep learning modelling techniques: current progress, applications, advantages, and challenges. Artif Intell Rev. 2023;56(11):13521–617. https://doi.org/10.1007/s10462-023-10466-8.
45. Tabane E, Mnkandla E, Wang Z. Ensemble deep learning models for multi-class DNA sequence classification: A comparative study of CNN, BILSTM, and GRU architectures. Appl Sci. 2026;16(3):1545. https://doi.org/10.3390/app16031545.
46. Ahmad N, Kumar V. A hybrid time series model for the spatio-temporal analysis of air pollution prediction based on pm 2.5. In: International conference on advanced network technologies and intelligent computing, Springer, 2023;62–81. https://doi.org/10.1007/978-3-031-64067-4_5.
47. Kour V, Kumar S, Misra R, Singh T. Advancing pm2. 5 and no 2 prediction accuracy through the convtrans model: A novel application of hybrid deep learning technologies. In 2024 IEEE International Conference on Big Data (BigData), 2024;4320–4326. IEEE. https://doi.org/10.1109/BigData62323.2024.10825906.
48. Ahmad N, Kumar V. Spatio-temporal forecasting using a hybrid bigru-1dcnn model for pm 2.5 concentrations in delhi, india (2018-2023) across multiple monitoring stations: N. ahmad, v. kumar. Water, Air, & Soil Pollution, 2025c;236(7):459. https://doi.org/10.1007/s11270-025-08103-x.
49. Kumar N. Atmospheric aerosols and urban chemistry in greater Noida. Sustain Chem Clim Action. 2026. https://doi.org/10.1016/j.scca.2026.100183.
50. Elminir HK. Dependence of urban air pollutants on meteorology. Sci Total Environ. 2005;350(1–3):225–37. https://doi.org/10.1016/j.scitotenv.2005.01.043.
51. Khanam Z, Sultana FM, Mushtaq F. Environmental pollution control measures and strategies: an overview of recent developments. Geospatial analytics for environmental pollution modeling: Analysis, control and management, 2023;385–414. https://doi.org/10.1007/978-3-031-45300-7_15.
52. Raju MP, Bhardwaj LK. Temporal trends in AGI: a case study of Noida. J Environ Sci Health. 2025;60(11):565–83. https://doi.org/10.1080/10934529.2025.2609042.
53. Schlesinger RB, Kunzli N, Hidy GM, Gotschi T, Jerrett M. The health relevance of ambient particulate matter characteristics: coherence of toxicological and epidemiological inferences. Inhalation Toxicol. 2006;18(2):95–125. https://doi.org/10.1080/08958370500306016.
54. Bhawan P, Nagar EA. Central pollution control board. Central Pollut. Control Board, New Delhi, India, Tech. Rep, 2020;20–21.
55. Zaini N, Ean LW, Ahmed AN, Malek MA. A systematic literature review of deep learning neural network for time series air quality forecasting. Environ Sci Pollut Res. 2022;29(4):4958–90. https://doi.org/10.1007/s11356-021-17442-1.
56. Kumar TV, Dahiya B. Smart economy in smart cities. In Smart Economy in Smart Cities: International Collaborative Research: Ottawa, St. Louis, Stuttgart, Bologna, Cape Town, Nairobi, Dakar, Lagos, New Delhi, Varanasi, Vijayawada, Kozhikode, Hong Kong, 2016;3–76. Springer. https://doi.org/10.1007/978-981-10-1610-3_1.

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